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POPULATION AND SETTLEMENT HISTORY OF THE TSARDOM OF RUSSIA, THE BALTIC, LITHUANIA-BELARUS, UKRAINE AND SIBERIA (1400-1721AD)

Regional Estimates of Population and Urban Settlement (1400-1721AD)

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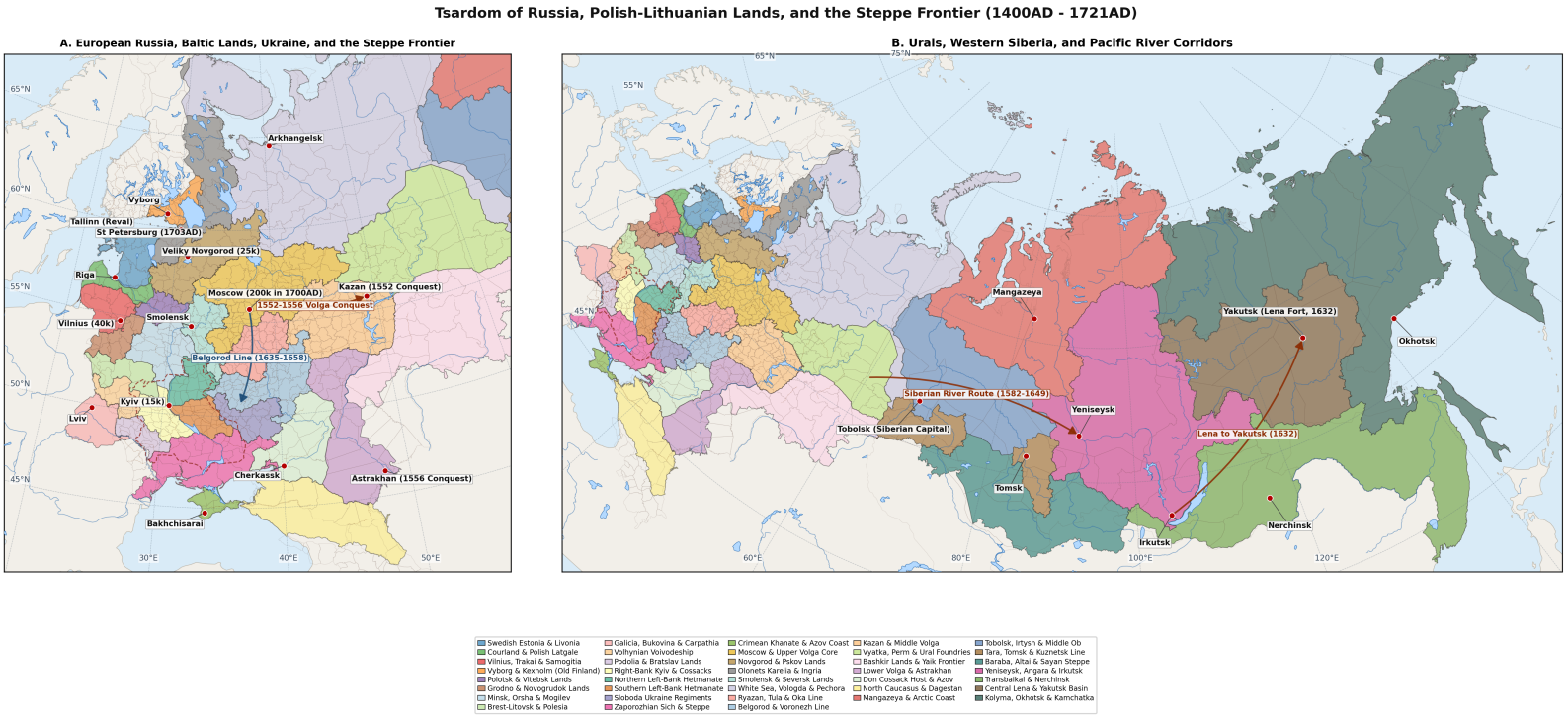


Figure 1. Historical regions and major urban centres across the Baltic provinces, Lithuania, Belarus, Ukraine, the Tsardom of Russia and Siberia (1400-1721AD).

Abstract

Between 1400AD and the Treaty of Nystad at the conclusion of the Great Northern War in 1721AD, the Northern Eurasian, East Slavic and Eastern Baltic lands underwent a profound demographic and institutional transformation. In 1400AD, the combined population across the thirty-eight historical regions spanning the Eastern Baltic confederations, the Grand Duchy of Lithuania, Vyborg-Kexholm Karelia, Crown and Cossack Ukraine, Crimea, the Tsardom of Russia, the North Caucasus and Siberia stood at 11,36M (11,36M). At that late medieval benchmark, regional population comprised 935k across Terra Mariana and Livonia, Courland-Inflanty, core Lithuania and Vyborg-Kexholm (Regions 1-4), 950k across the four Belarusian voivodeships of the Grand Duchy of Lithuania (Regions 5-8), 2,51M across the nine Ukrainian and Crimean-Pontic historical regions (Regions 9-17) and 6,96M across the twenty-one Russian, Volga-Ural, Steppe, Caucasian and Siberian regions (Regions 18-38, divided between 6,66M in

European Russia, the Steppe and the North Caucasus across Regions 18-30 and 300k across the eight Siberian zones of Regions 31-38) [1][2][3][4][5][6]. Over the subsequent three centuries, the total population doubled to 20,7M (20,7M) on the eve of the mid-17th-century wars in 1648AD and reached 22,83M (22,83M) at Peter I's First Poll-Tax Revision in 1720AD, comprising 1,26M in the Baltic, Lithuanian and Old Finland provinces (Regions 1-4), 1,55M in Belarus (Regions 5-8), 5,65M in Ukraine and Crimea (Regions 9-17) and 14,37M across Russia and Siberia (Regions 18-38, with 13,61M in Regions 18-30 and 760k in Regions 31-38) [7][8][9][10][11].

This long-term expansion was shaped by two opposing demographic forces. On one side, recurrent military, climatic and epidemic crises struck the western littoral and older forest cores: the Livonian War and Ivan IV's *Oprichnina* (1558-1583AD), the Huaynaputina volcanic-winter famine of 1601-1603AD and the Time of Troubles (*Smuta*, 1604-1618AD), the Russo-Polish War or Deluge (*Potop*, 1654-1667AD) across Lithuania and Belarus, *The Ruin* (*Ruina*, 1657-1687AD) in Right-Bank Ukraine, the Great Famine of 1695-1697AD (*Suur nalg* and *Suuret kuolonvuodet*) in Estland, Livland and Old Finland, and the Great Northern War and Baltic-Lithuanian Plague of 1708-1711AD [12][13][14][15][16]. On the other side, sustained frontier colonisation transformed the Left-Bank Cossack Hetmanate, Sloboda Ukraine, the Oka-Belgorod-Simbirsk fortification lines (*Zasechnye Cherty*), the Middle Volga, the Demidov Urals and the river-portage highways and southern fortified ploughlands of Siberia [17][18][19][20].

Background

Setting aside post-1991 state borders and excluding Central Asia and the South Caucasus prior to their 18th- and 19th-century annexations, this study divides the 18.119.929 km2 territory into thirty-eight historical regions based on Livonian and Swedish Baltic provinces, Courlandian and Polish-Lithuanian palatinates, Vyborg and Kexholm fiefs, Cossack regimental territories, Muscovite *uezds* and frontier *razryads*, Volga-Ural *darugas* and eight Siberian portage, ploughland, steppe and *yasak* zones (Figure 1 and Table 1). Across the Baltic, Lithuanian and Old Finland rim (Regions 1-4), Terra Mariana and Swedish Estland and Livland (Region 1, 76.229 km2) shifted from 260k in 1400AD to 360k in 1648AD and 300k in 1720AD; the Duchy of Courland and Semigallia and Polish Inflanty (Region 2, 37.284 km2) moved from 140k to 260k and 200k; the Grand Duchy of Lithuania core of Vilnius, Trakai and Samogitia (Region 3, 65.877 km2) expanded from 460k to 1,06M before contracting to 650k; and Vyborg Fief and Kexholm County (Region 4, 49.685 km2) rose from 75k to 110k. Across the four Belarusian voivodeships of the Grand Duchy of Lithuania (Regions 5-8), Polotsk and Vitebsk on the Western Dvina (Region 5, 32.576 km2, moving from 240k in 1400AD to 560k in 1648AD and 370k in 1720AD), Grodno and Novogrudok in Black Ruthenia (Region 6, 48.968 km2, from 280k to 740k and 490k), Minsk and the Orsha-Mogilev Dnieper corridor (Region 7, 102.278 km2, from 260k to 680k and 430k) and Brest-Litovsk and Pinsk-Turov Polesia (Region 8, 62.824 km2, from 170k to 520k and 260k) all peaked on the eve of the mid-17th-century wars before suffering heavy wartime losses.

Across the nine Ukrainian and Crimean-Pontic regions (Regions 9-17), the Ruthenian and Belz Voivodeships of Galicia with Bukovina and Carpathian Rus (Region 9, 71.558 km2) grew from 820k in 1400AD to 1,75M in 1648AD and 1,62M in 1720AD, whereas Volhynia (Region 10, 45.176 km2, moving from 420k to 920k and 680k), Podolia and Bratslav (Region 11, 39.620 km2, from 350k to 910k and 510k) and the Right-Bank Kyiv Voivodeship and Cossack regiments (Region 12, 56.764 km2, from 280k to 920k and 340k) experienced deep late-17th-century depopulation that fed rapid growth in the Northern Left-Bank Hetmanate around Chernihiv and Nizhyn (Region 13, 55.782 km2, from 210k to 490k and 790k), the Southern Left-Bank Hetmanate around Pereyaslav and Poltava (Region 14, 52.082 km2, from 110k to 460k and 760k) and the Sloboda Ukraine regiments around Kharkiv and Sumy (Region 15, 60.982 km2, from 50k to 160k and 400k), alongside the Zaporozhian Sich and Pontic Wild Fields (Region 16, 193.859 km2, from

60k to 130k and 180k) and the Crimean Khanate and Ottoman Eyalet of Kefe (Region 17, 32.002 km2, from 210k to 360k and 370k).

Within the thirteen European Russian, Volga-Ural, Steppe and North Caucasian regions (Regions 18-30), the Moscow and Zamoskovny core (Region 18, 277.154 km2) grew from 1,95M in 1400AD to 2,85M in 1648AD and 3,72M in 1720AD, flanked by the Novgorod and Pskov Five Pyatiny (Region 19, 159.354 km2, moving from 920k to 930k and 1,23M), Olonets Karelia and Baltic Ingria including Saint Petersburg (Region 20, 205.031 km2, from 200k to 220k and 350k), the Smolensk and Seversk borderlands (Region 21, 83.068 km2, from 480k to 640k and 730k) and the northern Pomorye, Vologda, Northern Dvina and Vychegda-Pechora lands (Region 22, 1.385.055 km2, from 210k to 460k and 420k). South and east of the Oka, agrarian and military colonisation transformed the Ryazan, Tula, Oryol and Great Oka Abatis Line (Region 23, 95.590 km2, from 680k to 1,08M and 1,52M), the Belgorod and Sevsk Razryads (Region 24, 148.778 km2, from 190k to 620k and 1,44M), the former Khanate of Kazan, Nizhny Novgorod and the Simbirsk-Penza Line (Region 25, 237.582 km2, from 780k to 1,18M and 1,76M), Great Perm, Vyatka and the Stroganov-Demidov Urals (Region 26, 479.789 km2, from 220k to 480k and 680k), the Bashkir Four Darugas and Yaik frontier (Region 27, 484.827 km2, from 180k to 260k and 340k), the Lower Volga, Astrakhan and Kalmyk Steppe (Region 28, 251.397 km2, from 240k to 320k and 430k), the Don Cossack Host and Azov (Region 29, 169.534 km2, from 90k to 165k and 250k) and the North Caucasus across Circassia, Kabarda, Vainakh societies, Dagestan and Terskiy Gorod (Region 30, 263.210 km2, from 520k to 665k and 740k). Across Northern Asia, the eight Siberian zones (Regions 31-38) comprised the Pomor-Mangazeya maritime and Taz-Turukhansk Arctic route (Region 31, 1.760.905 km2, from 16k to 26k and 18k), the Tobolsk-Irtysh-Middle Ob water highway (Region 32, 871.344 km2, from 42k to 88k and 158k), the southern Tara-Tomsk-Kuznetsk fortified agricultural frontier (Region 33, 359.192 km2, from 38k to 86k and 254k), the nomadic Baraba-Altai-Minusinsk-Tuva steppe borderlands (Region 34, 821.185 km2, from 74k to 84k and 100k), the Ket-Yeniseysk-Angara-Ilimsk-Irkutsk portage spine (Region 35, 1.931.504 km2, from 28k to 44k and 98k), Transbaikal Dauria and the Upper Amur frontier (Region 36, 1.753.934 km2, from 34k to 40k and 62k), the central Lena-Vilyuy-Aldan Yakutsk basin (Region 37, 1.373.412 km2, from 38k to 42k and 47k) and the Northeast Arctic and Pacific *yasak* frontier (Region 38, 3.924.538 km2, from 30k to 25k and 23k).

Historical Region/ Subtotal	**1400AD**	**1500AD**	**1550AD**	**1580AD**	**1600AD**	**1615AD**	**1648AD**	**1667AD**	**1678AD**	**1700AD**	**1720AD**
1. Terra Mariana/ Swedish Estland & Livland	260	330	390	270	290	260	360	340	390	440	300
2. Courland, Semigallia & Polish Inflanty	140	175	210	190	210	200	260	210	240	280	200
3. GDL Core: Vilnius, Trakai & Samogitia	460	610	720	780	850	890	1.060	640	710	810	650

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Historical Region/ Subtotal	1400AD	1500AD	1550AD	1580AD	1600AD	1615AD	1648AD	1667AD	1678AD	1700AD	1720AD
4. Vyborg Fief & Kexholm/ Old Finland	75	92	105	88	95	80	110	105	118	130	110
5. Polotsk & Vitebsk Voivode-ships (GDL)	240	320	390	350	410	440	560	270	310	360	370
6. Grodno & Novo-grudok Voivode-ships (GDL)	280	385	470	520	580	620	740	420	470	520	490
7. Minsk, Orsha & Mogilev Dnieper (GDL)	260	350	430	460	510	550	680	330	380	430	430
8. Brest-Litovsk & Pinsk-Turov Polesia (GDL)	170	245	310	340	380	410	520	260	290	320	260
9. Ruthenian/ Galicia, Belz & Carpathian Rus	820	1.020	1.180	1.290	1.420	1.510	1.750	1.380	1.450	1.580	1.620
10. Volhynian Voivode-ship (Lutsk & Ostroh)	420	530	620	680	750	790	920	580	610	660	680
11. Podolian & Bratslav Voivode-ships	350	450	540	610	690	740	910	490	380	440	510

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Historical Region/ Subtotal	1400AD	1500AD	1550AD	1580AD	1600AD	1615AD	1648AD	1667AD	1678AD	1700AD	1720AD
12. Right-Bank Kyiv Regiments (Chyhyryn)	280	360	440	520	620	690	920	520	360	320	340
13. N. Left-Bank Hetmanate (Chernihiv-Nizhyn)	210	260	300	330	370	390	490	560	620	720	790
14. S. Left-Bank Hetmanate (Poltava-Lubny)	110	145	180	220	270	320	460	540	610	710	760
15. Sloboda Ukraine Regiments (Kharkiv-Sumy)	50	65	80	95	110	120	160	250	310	380	400
16. Zaporozhian Sich & Pontic Wild Fields	60	70	80	90	100	105	130	135	145	165	180
17. Crimean Khanate & Ottoman Kefe	210	260	290	310	330	335	360	365	375	385	370
18. Moscow & Zamoskovny Core (Vladimir-Tver)	1.950	2.620	3.050	2.780	2.950	2.280	2.850	3.080	3.320	3.620	3.720
19. Novgorod & Pskov Five Pyatiny	920	1.170	1.340	960	1.040	710	930	1.010	1.090	1.180	1.230

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Historical Region/ Subtotal	1400AD	1500AD	1550AD	1580AD	1600AD	1615AD	1648AD	1667AD	1678AD	1700AD	1720AD
20. Olonets Karelia & Baltic Ingria (St P.)	200	250	280	220	240	180	220	230	250	270	350
21. Smolensk & Seversk Border-lands (Bryansk)	480	600	680	640	690	510	640	580	640	710	730
22. Pomorye, Vologda, Dvina & Pechora North	210	280	340	380	410	420	460	470	480	450	420
23. Ryazan, Tula, Oryol & Great Oka Abatis	680	850	980	940	1.020	820	1.080	1.180	1.290	1.420	1.520
24. Belgorod & Sevsk Razryads (Voronezh-Kursk)	190	240	290	340	410	380	620	820	980	1.220	1.440
25. Kazan, Nizhny Novgorod & Simbirsk Line	780	890	940	960	1.020	980	1.180	1.320	1.460	1.620	1.760
26. Vyatka, Great Perm & Stroganov Urals	220	280	330	360	390	405	480	520	560	610	680
27. Bashkir Four Darugas (Ufa) & Yaik Frontier	180	200	215	225	235	240	260	275	290	310	340

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Historical Region/ Subtotal	1400AD	1500AD	1550AD	1580AD	1600AD	1615AD	1648AD	1667AD	1678AD	1700AD	1720AD
28. Lower Volga, Astrakhan & Kalmyk Steppe	240	260	270	265	275	280	320	350	375	400	430
29. Don Cossack Host (Cherkassk) & Azov	90	100	110	120	135	130	165	190	215	240	250
30. North Caucasus (Circassia, Kabarda, Dagestan)	520	565	595	610	625	635	665	680	695	715	740
31. Pomor-Mangazeya Maritime & Taz Portage	16	17	18	19	24	28	26	22	19	18	18
32. Tobolsk-Irtysh-Middle Ob Portage Highway	42	45	47	49	56	64	88	105	118	138	158
33. S. Tobol-Tara-Tomsk-Kuznetsk Ploughland	38	42	45	47	52	60	86	118	145	195	254
34. Nomadic Baraba-Altai-Minusinsk Steppe	74	77	80	81	82	82	84	82	80	88	100
35. Ket-Yeniseysk-Angara-Ilimsk-Irkutsk Spine	28	30	31	32	33	34	44	56	66	82	98

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Historical Region/ Subtotal	1400AD	1500AD	1550AD	1580AD	1600AD	1615AD	1648AD	1667AD	1678AD	1700AD	1720AD
36. Trans-baikal Dauria & Upper Amur (Nerchinsk)	34	36	37	38	39	39	40	44	48	55	62
37. Central Lena-Vilyuy-Aldan (Yakutsk Alaas)	38	40	41	42	42	41	42	43	44	45	47
38. NE Arctic & Pacific Yasak (Kolyma-Kamchatka)	30	30	31	30	29	27	25	25	25	24	23
Baltic, Lithuania & Old Finland (Reg. 1-4)	935	1.207	1.425	1.328	1.445	1.430	1.790	1.295	1.458	1.660	1.260
Belarus GDL Subtotal (Reg. 5-8)	950	1.300	1.600	1.670	1.880	2.020	2.500	1.280	1.450	1.630	1.550
Ukraine & Crimea Subtotal (Reg. 9-17)	2.510	3.160	3.710	4.145	4.660	5.000	6.100	4.820	4.860	5.360	5.650
Russia European & Steppe (Reg. 18-30)	6.660	8.305	9.420	8.800	9.440	7.970	9.870	10.705	11.645	12.765	13.610
Russia Siberian Zones (Reg. 31-38)	300	317	330	338	357	375	435	495	545	645	760
Russia Subtotal (Reg. 18-38)	6.960	8.622	9.750	9.138	9.797	8.345	10.305	11.200	12.190	13.410	14.370

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Historical Region/ Subtotal	1400AD	1500AD	1550AD	1580AD	1600AD	1615AD	1648AD	1667AD	1678AD	1700AD	1720AD
Total Domain (All 38 Regions)	11.355	14.289	16.485	16.281	17.782	16.795	20.695	18.595	19.958	22.060	22.830

Table 1. Regional population estimates and polity subtotals across all thirty-eight historical regions (1400-1720AD).

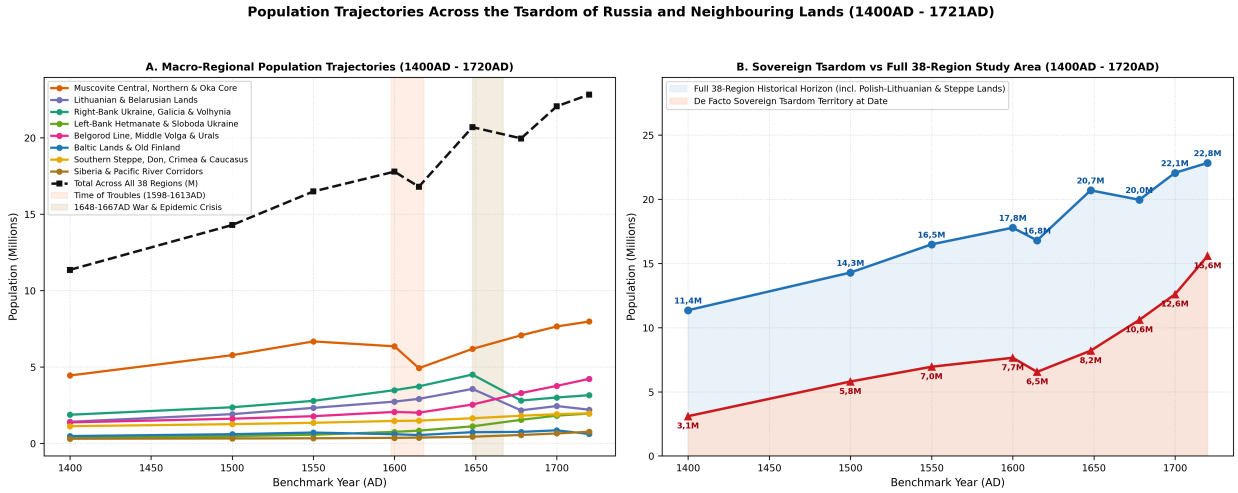


Figure 2. Regional population trajectories across Russia, Ukraine, Belarus, Lithuania and the Baltic provinces (1400-1720AD).

Methodology

Demographic estimates across the thirty-eight regions between 1400AD and 1721AD rest on six complementary fiscal and ecclesiastical accounting traditions that gradually replaced indirect arable-unit assessments with household, hearth and individual soul enumerations [3][7][21][22]. In the Muscovite core and northern provinces (Regions 18-26), late 15th- and 16th-century registration relied on the *pistsovye knigi* (scribe land cadastres), which assessed obligations in units of arable land known as the *sokha* (400 to 800 *chetverti* of good arable land in three fields, scaled by service, church or state-peasant tenure) and the Novgorodian *obzha* (Regions 19-20) [2][23]. Because peasants reacted to wartime fiscal pressure by abandoning assessed ploughland or merging nuclear families into extended households, cadastral *dvor* counts require calibration against observed household multipliers of 6 to 8,5 persons [1][19].

The shift from land-area (*pozemelnoe*) to household (*podvornoe*) taxation in the Tsardom of Russia produced the general enumerations of 1646AD and 1678AD (*perepisnye knigi*) [7][21]. Cross-checking the 1678AD household books against the 1710AD wartime audit and Peter I's First Poll-Tax Revision (1719-1721AD, audited by military *popolnenie* commissions in 1721-1724AD), Ya. E. Vodarsky (1977) and V. M. Kabuzan (1963) showed that pre-Petrine household registers omitted rural clergy, *sluzhilye lyudi*

servicemen and untaxed *bobyli* or *zadvornye* dependents while encouraging household amalgamation to 8,5-9,2 persons (4,2-4,6 males) per *dvor* [7][8]. Replacing the household levy with the individual male capitation tax (*podushnaya podat*) in 1718-1719AD removed the fiscal incentive to conceal split families and registered 7,18M male souls (14,37M total inhabitants of both sexes across Regions 18-38 in 1720AD) [8][24].

Along the western Baltic and Fennoscandian littoral (Regions 1, 2 and 4), demographic series are anchored by Livonian and Swedish *Hakenrevisionen* (ploughland *adramaa* and *haken* audits of 1586-1591AD, 1601AD, 1638AD, 1688AD and 1712AD), Lutheran parish registers (*kirikuraamatud* introduced across Estland, Livland and Vyborg-Kexholm from the mid-17th century), Courlandian ducal domain inventories and Swedish poll-tax (*mantalslangder*) rolls reconstructed by H. Palli (1996), S. Vahtre (1973), E. Dunsdorfs (1962) and V. Saloheimo (1976) [3][10][11][25]. In core Lithuania (Region 3), Belarus (Regions 5-8) and Polish Crown Ukraine (Regions 9-12), estimates rest on the 1528AD Grand Duchy of Lithuania Military Census (*Popis voyska*, recording 19.842 noble cavalrymen at a statutory ratio of 1 horseman per 8 peasant *sluzhby* alongside grand-ducal and church estates), the 1557AD *Voloka* agrarian cadastres (*Pomiara wloczna*), Crown royal *lustracje* (1564-1565AD, 1616AD and 1629AD) and *podymne* hearth-tax registers of 1629AD, 1648-1650AD, 1667AD, 1690AD and 1717AD [4][5][9][13][26]. Finally, the Cossack Hetmanate and Sloboda Ukraine (Regions 12-15), Crimea (Region 17) and the eight Siberian zones (Regions 31-38) draw on the 1649AD Zboriv Register, the 1666AD Left-Bank *perepysni knyhy*, Ottoman *tahrir defterleri* of Kefe and Siberian *yasachnye* and *dozornye knigi* [6][20][27][28].

Documentary Corpus	Period	Primary Accounting Unit	Demographic Multiplier	Regional Coverage	Principal Scholarship
Baltic Hakenre-visionen & Mantalslangder	1561-1715AD	Adramaa/ Haken, farmstead (talu) & parish soul	7,5-9,5 persons/talu farmstead	Regions 1, 2 & 4 (Estland, Livland, Courland, Vyborg)	Palli 1996; Vahtre 1973; Dunsdorfs 1962; Saloheimo 1976
Lithuanian 1528AD Popis & Voloka Books	1528-1600AD	Cavalry horse, sluzhba & voloka (21,36 ha)	6,5-7,5 persons/ voloka-sluzhba	Regions 3, 5-8 & 10 (Lithuania, Belarus & Volhynia)	Lowmianski 1983; Ochmanski 1982; Vasiliauskas 2012
Polish Crown Lustracje & Podymne Hearths	1564-1717AD	Taxable hearth (dym) & lan arable unit	6-6,5 persons/ rural dym	Regions 2-3 & 5-12 (Inflanty, GDL & Crown Rus)	Morzy 1965; Karpachev & Kozlovsky 1972; Yakovenko 2006
Novgorod & Moscow Pistsovye Knigi	1495-1594AD	Sokha, obzha & peasant household (dvor)	5,5-7 persons/ dvor	Regions 18-23 (Zamoskovny, Pyatiny, Ingria, Oka)	Kopanev 1959; Shapiro 1971, 1974; Selin 2020
Zboriv Register, 1666AD Books & Tahrir	1520-1700AD	Cossack kurin/ dvor, posad dvor & Ottoman hane	5,5-7 persons/ household	Regions 12-17 (Hetmanate, Sloboda & Crimea)	Krypiakevych 1954; Irtenina 2022; Ozturk 2019
1646AD & 1678AD Perepisnye Knigi	1646AD; 1678AD	Taxable household (tyaglyy dvor)	8-9,2 persons/ enlarged dvor	Regions 18-29 (Muscovite Core & Frontier Razryads)	Vodarsky 1973, 1977; Avdeev 2025

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Documentary Corpus	Period	Primary Accounting Unit	Demographic Multiplier	Regional Coverage	Principal Scholarship
Siberian Yasachnye & Dozornye Knigi	1595-1719AD	Yasak-paying adult male (18-50 years)	4-4,2 total pop/ yasak payer	Regions 31-38 (8 Siberian Portage & Yasak Zones)	Dolgikh 1960; Pokrovsky 1984; Ushnitskiy 2017b
First Poll-Tax Revision (Revizskie Skazki)	1719-1724AD	Individual male soul (revizskaya dusha)	2-2,02 total pop/male soul	Regions 1, 4, 13-15 & 18-38 (Petrine Empire)	Kabuzan 1963, 1990; Ulyanova 2016; Mironov 2007

Table 2. Principal fiscal cadastres, ploughland audits, parish registers and household censuses used for regional population estimates (1400-1721AD).

Results

The Baltic, Lithuania and Old Finland

Estland, Livland and Courland

Across the Eastern Baltic littoral (Regions 1 and 2, covering 113.513 km2 combined), late medieval political authority belonged to the ecclesiastical and knightly confederation of *Terra Mariana* (the Archbishopric of Riga, the Bishoprics of Dorpat, Osel-Wiek and Courland, and the Livonian Branch of the Teutonic Order) alongside Hanseatic mercantile municipalities led by Riga (8k in 1400AD to 12k in 1550AD), Reval/Tallinn (5,5k to 8k), Dorpat/Tartu (4,5k to 6,5k), Pernau/Parnu (1,6k to 2,6k), Narva (1,8k to 3,2k), Wenden/Cesis (1,5k to 2,4k) and Arensburg/Kuressaare [3][10][29]. Favoured by a century of relative internal peace after 1435AD and rising rye, flax and timber exports to Western Europe, Terra Mariana/Estland and Livland (Region 1, 76.229 km2) grew from 260k in 1400AD to 330k in 1500AD and 390k in 1550AD, while southern Courland, Semigallia and Latgale (Region 2, 37.284 km2) rose from 140k to 210k [3][10][25].

Ivan IV's invasion of Narva and Dorpat in 1558AD shattered the medieval confederation and initiated seventy years of intermittent Baltic warfare. During the Livonian War (1558-1583AD) and the subsequent Polish-Swedish wars of 1600-1629AD - compounded by the extreme volcanic-winter crop failure, famine and plague of 1601-1603AD - rural *adramaa* farmstead occupancy across mainland Estland and northern Livland dropped by more than half [3][14]. Region 1 fell by 30,8% between 1550AD (390k) and 1580AD (270k) and stood at 260k in 1615AD before the Truce of Altmark (1629AD) and Treaty of Bromsebro (1645AD, ceding Osel/Saaremaa from Denmark) consolidated Swedish Estland and Livland [3][30]. Under 17th-century Swedish administration, inward rural migration of Finnish, Russian, Latvian and Lithuanian settlers, parish schooling and Charles XI's *Reduktion* of noble estates in the 1680sAD drove demographic recovery. Region 1 climbed to 360k in 1648AD, 390k in 1678AD and 440k at the turn of the century (c. 1695-1700AD), despite the severe Great Famine of 1695-1697AD (*Suur nalg*), when cold, rain-soaked summers destroyed two consecutive harvests and killed roughly 70k-75k inhabitants across Estland and Livland while Swedish authorities continued shipping Baltic manor grain to Stockholm and Finland [3][14][31].

South of the Daugava (Western Dvina), the Pact of Vilnius (1561AD) secularised the southern Ordensstaat

into the hereditary Duchy of Courland and Semigallia under Gotthard Kettler as a fief of the Polish-Lithuanian Commonwealth, while eastern Latgale remained under direct Commonwealth sovereignty after 1629AD as the Voivodeship of Inflanty (Dunaburg/Daugavpils, Rezhitsa/Rezekne and Ludza) [10][32]. Shielded from the worst fighting of the Livonian War (dipping mildly from 210k in 1550AD to 190k in 1580AD), Region 2 expanded to 260k in 1648AD under the mercantilist manufactures, shipyards and Atlantic-Baltic trade of Duke Jacob Kettler (1642-1682AD) at his capital Mitau/Jelgava (5,5k in 1648AD), the shipbuilding port of Windau/Ventspils (3,2k), Libau/Liepaja (3,5k), Goldingen/Kuldiga (3k) and Bauske (2,5k) [10][33]. After suffering a 19,2% contraction (260k to 210k) during the 1658-1660AD Swedish occupation of Courland and Muscovite capture of Dunaburg (briefly renamed *Borisoglebsk* in 1656-1666AD), Region 2 recovered to 280k by 1700AD [10][34].

Both Baltic regions then suffered heavy losses during the Great Northern War (1700-1710AD) and the Great Baltic Plague of 1710-1711AD [3][15][33][34]. In Estland and Livland (Region 1), the population plunged by 43,2% in a single decade, from 440k in 1700AD to 250k in 1710AD, as Russian and Swedish campaigns around Narva (1700AD and 1704AD), Dorpat (besieged in 1704AD and demolished in 1708AD with its burghers deported to Vologda), Pernau and Riga culminated in the *Yersinia pestis* epidemic of summer-autumn 1710AD. That outbreak killed over half the besieged population of Riga (15k to 6,5k), roughly two-thirds of Reval (5k to 4,5k) and up to 70% of western Estland coastal parishes before the Baltic knighthoods signed the Capitulations of 1710AD, confirmed by the Treaty of Nystad in 1721AD as the regional total recovered to 300k in 1720AD [3][15][30]. Across Courland, Semigallia and Polish Inflanty (Region 2), the population fell by 39,3%, from 280k in 1700AD to 170k in 1710AD, before recovering to 200k by 1720AD; Saxon, Swedish and Russian armies fought across the Daugava (1701AD) and Gemauerthof (1705AD), and the 1710AD plague wiped out nearly half of Courland's rural inhabitants while reducing Mitau from 6,8k to 2,6k [10][33][34].

Core Lithuania

Immediately south of Courland and Livonia lay the political and ethnic Lithuanian core of the Grand Duchy of Lithuania in Region 3 (65.877 km2), uniting the western districts of the Vilnius and Trakai Voivodeships with the Eldership of Samogitia (*Zemaitija*) [4][35]. Untouched by Tatar steppe raiding north of the Neman (Nemunas) and stimulated by German/Prussian and Baltic river commerce through the Hanseatic factory at Kaunas, Region 3 grew steadily from 460k in 1400AD to 610k in 1500AD and 720k in 1550AD. In Henryk Lowmianski's and Jerzy Ochmanski's reconstructions of the 1528AD GDL Military Census (*Popis voyska*), ethnic Lithuania and Samogitia exhibited the highest rural settlement densities in the entire Grand Duchy [4][26]. Following Sigismund II Augustus's 1557AD Volok Reform (*Valaku reforma*), which reorganised grand-ducal and magnate domains (such as the Siauliai Crown Economie) into regular three-field *voloka* villages and chartered scores of market towns (*miesteliai*), Region 3 expanded to 850k in 1600AD and 1,06M in 1648AD [35][36]. At its mid-17th-century peak, the Grand-Ducal capital Vilnius (seat of the 1579AD Jesuit University, the Lithuanian Tribunal and thriving Ruthenian, Polish, Lithuanian, Jewish, German and Karaite communities) reached ~28k inhabitants, flanked by Kaunas (~15k), the Radziwill Protestant commercial hub of Kedainiai (~6,5k), Trakai (~5k), Siauliai (~3,5k), Raseiniai (~3,2k, parliamentary seat of Samogitia) and Telsiai (~2,6k) [35][36][37].

This growth ended in August 1655AD when Tsar Aleksey Mikhailovich's armies and Ivan Zolotarenko's Cossacks captured Vilnius - the first foreign capture of the Lithuanian capital in its history - sparking a seventeen-day conflagration, while Swedish forces occupied Samogitia and Kedainiai [13][35]. Six years of Muscovite occupation in Vilnius and Trakai (1655-1661AD), famine, the 1657-1658AD plague and artisan deportations reduced Region 3 by 39,6%, from 1,06M in 1648AD to 640k in 1667AD (with Vilnius shrinking to 11,5k and Kaunas to 6k) [4][13]. Although post-war reconstruction lifted Region 3 back to 710k in 1678AD and 810k in 1700AD, the Great Northern War (during which Swedish, Russian, Saxon and rival

Sapieha-Oginski Lithuanian confederate armies ravaged Samogitia and Vilnius) and the Great Plague of 1708-1711AD killed more than a quarter of the population, driving Region 3 down to 580k in 1710AD (-28,4%) before an initial recovery to 650k in 1720AD [16][32][35].

Vyborg and Kexholm

On the north-western Karelian Isthmus, Lake Saimaa and the western shore of Lake Ladoga, Region 4 (49.685 km2) bridges the Swedish-Finnish and Novgorodian-Russian worlds [11][38]. Throughout the 15th and early 16th centuries, the western half of this territory formed the Swedish frontier province of Vyborg Fief (*Viipurin linnalaani*), guarded by the stone castle of Vyborg/Viipuri (chartered 1403AD, 2,2k in 1400AD to 3,6k in 1550AD) and the inland Savo lake fortress of Nyslott/Savonlinna (*Olavinlinna*, founded 1475AD), while the eastern Ladoga shore belonged to Novgorod's and Muscovy's Korela (*Kexholm/Priozersk*) uezd [23][38]. Combined population grew from 75k in 1400AD to 105k in 1550AD before the 25-year Russo-Swedish "Long Wrath" (*Pitka viha*, 1570-1595AD) and Swedish occupation of Kexholm (1580-1595AD) depressed regional numbers to 88k in 1580AD and 80k during the *Smuta* and the Ingrian War in 1615AD [38][39].

When the Treaty of Stolbovo (February 1617AD) transferred Kexholm County (*Kakisalmen laani*) to the Swedish Empire, Region 4 underwent a near-total ethno-confessional replacement documented in Swedish tax rolls and Muscovite *pistsovye knigi* by V. Saloheimo (1976), K. Katajala (2017), O. M. Fishman (2020) and Yu. V. Stepanova (2024) [11][38][40][41]. Facing heavy Swedish crown rents, noble fief (*forlaning*) exactions and Lutheran proselytising under Bishop Isaac Rothovius's successors, Orthodox Karelian families migrated *en masse* across the border into Russia - first during the 1620s-1640sAD (some 25k migrants) and decisively during the Russo-Swedish War of 1656-1658AD (*Ruptuuri*), when an additional 20k-25k Orthodox Karelians withdrew with Russian troops and were settled by Tsar Aleksey Mikhailovich on crown lands in Bezhetskiy Verkh, Tver, Torzhok, Vyshny Volochyok and Valdai (Regions 18-19), creating the Tver Karelian population [40][41]. Simultaneously, Lutheran peasants from Savonia (*Savo*) and Vyborg Karelia poured into the vacated Kexholm parishes to clear slash-and-burn (*kaski*) rye plots, lifting Region 4 to 110k in 1648AD and 130k by 1700AD alongside newly chartered Saimaa-Gulf of Finland tar-export towns at Villmanstrand/Lappeenranta (1649AD) and Vehkalahti/Fredrikshamn (Hamina, 1653AD) [11][38]. The Finnish Great Famine of 1696-1697AD (*Suuret kuolonvuodet*), followed by Peter I's 1710AD Sieges of Vyborg and Kexholm (conceived as the northern security glacis for Saint Petersburg) and the Great Wrath occupation (*Isoviha*), cut Region 4 by 26,9% to 95k in 1710AD before recovering to 110k in 1720AD upon formal Russian annexation of Vyborg and Kexholm ("Old Finland", *Vanha Suomi*) at the Treaty of Nystad (1721AD) [11][38].

Western/Baltic Region	Area	Principal Urban & Fortress Centres	1648AD/ 1700AD	Crisis Trough (% Drop)	1720AD
1. Estland & Livland (Terra Mariana/Ostsee)	76.229 km2	Riga, Reval (Tallinn), Dorpat (Tartu), Narva, Pernau, Wenden, Arensburg	360k/440k	250k (1710AD, -43,2%)	300k
2. Courland, Semigallia & Polish Inflanty	37.284 km2	Mitau (Jelgava), Libau, Windau, Goldingen, Bauske, Dunaburg, Rezhitsa	260k/280k	170k (1710AD, -39,3%)	200k
3. GDL Core: Vilnius, Trakai & Samogitia	65.877 km2	Vilnius, Kaunas, Trakai, Kedainiai, Siauliai, Raseiniai, Telsiai	1,06M/ 810k	640k (1667AD)/ 580k (1710AD)	650k

(Continued from previous page)

Western/Baltic Region	Area	Principal Urban & Fortress Centres	1648AD/ 1700AD	Crisis Trough (% Drop)	1720AD
4. Vyborg Fief & Kexholm/Old Finland	49.685 km2	Vyborg (Viipuri), Kexholm (Priozersk), Nyslott, Fredrikshamn, Villmanstrand	110k/130k	80k (1615AD)/ 95k (1710AD)	110k
Western Rim Subtotal (Regions 1-4)	229.075 km2	27 chartered Hanseatic, Magdeburg, ducal and Swedish staple towns	1,79M/ 1,66M	1,09M (1710AD, -34% vs 1700AD)	1,26M

Table 3. Population summary of the Baltic provinces, Lithuania and Old Finland (1400-1720AD), showing pre-war peaks, epidemic troughs and 1720AD totals.

The Muscovite Core and the Russian North

The Forest Heartland, Novgorod and Pomorye

At the opening of the 15th century, the five northern and western forest regions of Russia (Regions 18-22, totalling 3,76M in 1400AD) were recovering from recurrent waves of the Black Death (1352-1353AD, 1364-1366AD, 1417AD and 1425-1427AD) and the destructive campaigns of Tokhtamysh (1382AD) and Edigu (1408AD) [1][42]. Within the Moscow and Zamoskovny core (Region 18, 277.154 km2, rising from 1,95M in 1400AD to 3,05M in 1550AD), internal agricultural colonisation advanced along the watershed ridges (*vodorazdely*) of the Moscow-Klyazma basin, the fertile Suzdal *Opolye*, Tver, Yaroslavl, Kostroma and Kaluga, led by peasant clearance and monastic settlement networks around the Trinity-Sergius Lavra [2][23].

Under Ivan III (1462-1505AD) and Vasily III (1505-1533AD), Moscow annexed the Novgorod Republic (1478AD), the Grand Principality of Tver (1485AD), Vyatka (1489AD), the Upper Oka and Seversk principalities (1494-1503AD), Pskov (1510AD), Smolensk (1514AD) and Ryazan (1521AD) [43][44][45]. Because the Ilmen-Pskov arable heartland and the Finno-Karelian and Neva borderlands followed distinct institutional and military trajectories between 1478AD and 1721AD, the regional framework separates the Novgorod and Pskov lands (Region 19) from Olonets Karelia and Baltic Ingria (Region 20), while western Vyborg and Ladoga Kexholm are tracked in Region 4.

The Novgorod and Pskov lands (Region 19, 159.354 km2) grew from 920k in 1400AD to 1,17M in 1500AD and 1,34M in 1550AD across the Shelonskaya, Derevskaya, Bezhetskaya and southern Vodskaya and Obonezhskaya *pyatiny* around Lake Ilmen, the Volkhov and the Lovat, together with the Pskov Republic (Pskov, Izborsk, Ostrov, Opochka and Gdov) and the southern border fortress of Velikie Luki [23][44]. Following the confiscation of Novgorodian boyar and archiepiscopal estates in 1478-1489AD and the settlement of Moscow cavalry *pomeshchiki*, the surviving *pistsovye knigi* of 1495-1505AD and the 1540s-1550sAD document a 14,5% demographic expansion between 1500AD and 1550AD, with Veliky Novgorod (~33k-35k in 1500-1550AD) and Pskov (~28k-30k inside the Okolny Gorod wall) ranking as the second and third cities of the Tsardom [12][23].

To the north, Olonets Karelia and Baltic Ingria (Region 20, 205.031 km2) moved from 200k in 1400AD to 280k in 1550AD, 180k in 1615AD and 350k in 1720AD. This territory spanned the Neva and Gulf of Finland littoral (*Izhora/Ingria*, including Ivangorod founded in 1492AD opposite Narva, Yam, Koporye and Oreshek/Noteburg at the Neva outlet) together with Olonets-Onega Karelia (*Zaonezhskie* and *Lopatinskie*

pogosty, with Olonets fortress founded in 1649AD) [23][39]. Unlike the Ilmen-Pskov grain belt, Region 20 combined slash-and-burn (*podseka*) agriculture, Ladoga-Onega fisheries, bog-iron smelting (*domnitsy*, and later the Petrine Olonets ironworks at Petrovskaya Sloboda in 1703AD) and the Neva-Ingrian fortresses ceded to Sweden under the Treaty of Stolbovo (1617AD) before Peter I's reconquest and the foundation of Saint Petersburg in 1703AD [12][39].

Along the western frontier, the Smolensk and Seversk borderlands (Region 21, 83.068 km2, moving from 480k in 1400AD to 680k in 1550AD and 730k in 1720AD) guarded the Upper Dnieper and Desna-Bryansk corridor between Moscow and the Grand Duchy of Lithuania, anchored by Fyodor Kon's brick citadel at Smolensk (1595-1602AD), Dorogobuzh, Vyazma, Roslavl and Bryansk [1]. Across the Russian North, Pomorye, Vologda, the Northern Dvina and the Vychegda-Pechora Komi lands (Region 22, 1.385.055 km2, moving from 210k in 1400AD to 340k in 1550AD, 480k in 1678AD and 420k in 1720AD) formed a *chernososhnye* (state-peasant) maritime, salt-boiling, fur-trading and Zyryan Komi region. After the English navigator Richard Chancellor reached the Northern Dvina in 1553AD, the White Sea trade route stimulated urban growth along the Sukhona-Dvina axis linking Vologda (13k in 1550AD), Veliky Ustyug (5k), Solvychegodsk (4k), Kholmogory (8k), Arkhangelsk (founded 1584AD), Kargopol, Kola and the Pechora portage outpost of Pustozersk (1499AD) [1][42]. Region 22 peaked at 480k in 1678AD before declining to 420k by 1720AD as Petrine military conscription, naval timber levies and the 1713-1718AD decrees diverting foreign maritime trade from Arkhangelsk to Saint Petersburg (Region 20) drew population out of the White Sea basin [7].

The Oprichnina and the Livonian War

The demographic expansion of the old Muscovite heartland reversed during the second half of the reign of Ivan IV (1547-1584AD), when four overlapping shocks struck Regions 18, 19, 20 and 21 between 1565AD and 1583AD [12][39][46]. Twenty-five years of mobilisation during the Livonian War (1558-1583AD) and extraordinary wartime levies (*pyatinnye dengi*, *yam* post obligations and siege-artillery transport) exhausted the tax-paying peasantry (*tyaglye krestyane*) and *pomeshchik* service estates across the Novgorod, Pskov, Ingrian, Smolensk and Zamoskovny districts [12][39]. At the same time, Ivan IV's division of the realm into the *Zemshchina* and the *Oprichnina* (1565-1572AD) brought estate confiscations, heavy exactions and the punitive expedition against Tver, Torzhok, Klin and Veliky Novgorod in January-February 1570AD, killing thousands of inhabitants and disrupting the regional grain trade [1][12].

Environmental and military disasters compounded the fiscal crisis. Early frosts and harvest failure in 1569-1570AD triggered severe famine across central and north-western Russia, accompanied by a lethal plague epidemic (*morovoe povetrie*) throughout Regions 18-20 [12]. In May 1571AD, Khan Devlet I Giray bypassed the Oka defences (Region 23) and burned the wooden suburbs of Moscow (Region 18), killing tens of thousands and carrying off an estimated 60k-100k captives before Prince Mikhail Vorotynsky defeated the Crimean host at the Battle of Molodi in 1572AD [47][48].

The cadastral books compiled in the 1580sAD and early 1590sAD document widespread rural abandonment (*zapustenie*) [1][12]. In Novgorod and Pskov (Region 19), between 75% and 90% of previously cultivated *obzhi* were registered as deserted wasteland (*pustoshi* or *porozhnie zemli*); regional population dropped by 28,4%, from 1,34M in 1550AD to 960k in 1580AD, while Olonets Karelia and Baltic Ingria (Region 20) fell by 21,4% (280k to 220k) during Pontus De la Gardie's Swedish capture of Ivangorod, Yam and Koporye (1581-1590AD, recovered by Russia at the 1595AD Truce of Teusina) [12][39]. The Moscow and Zamoskovny core (Region 18) contracted from 3,05M (1550AD) to 2,78M (1580AD), and Smolensk and Seversk (Region 21) declined from 680k to 640k [1][45]. Conversely, fugitive peasants moved north into Pomorye (Region 22, 340k to 380k), south into the Belgorod and Sevsk Razryads (Region 24, 290k to 340k) and east into Kazan and the Middle Volga (Region 25, 940k to 960k) and Perm-Urals (Region 26, 330k to 360k), prompting the Muscovite crown to introduce the *zapovednye leta* ("forbidden years",

from 1581AD) suspending peasant departure on St George's Day (*Yuryev den*) [19][46].

Belarus and Ukraine Before 1648AD

The Belarusian Voivodeships

In contrast to late 16th-century Muscovy, the four Belarusian regional clusters of the Grand Duchy of Lithuania (Regions 5-8, covering 246.646 km2) experienced sustained demographic and urban expansion between 1400AD and 1648AD, with the combined Belarusian population rising from 950k in 1400AD to 1,3M in 1500AD, 1,6M in 1550AD, 1,88M in 1600AD and 2,5M on the eve of the mid-17th-century wars in 1648AD [9][13][26]. In the Polotsk and Vitebsk Voivodeships (Region 5, 32.576 km2), anchored by the Western Dvina export artery to Riga and the commercial cities of Polotsk (~20k in 1648AD), Vitebsk (~17k), Drissa and Lepel, the population expanded from 240k in 1400AD to 390k in 1550AD, dipped temporarily to 350k in 1580AD during Ivan IV's occupation of Polotsk (1563-1579AD) and Stephen Bathory's reconquest, and climbed to 560k in 1648AD [13][49]. Across the Grodno and Novogrudok Voivodeships (Region 6, 48.968 km2), comprising Black Ruthenia (*Chornaya Rus*) and the fertile Ponemanje basin around Grodno (~13,5k), Novogrudok (~9k), Slonim (~7,2k), Lida (~6k) and Volkovysk (~5,5k), the population grew from 280k in 1400AD to 470k in 1550AD and 740k in 1648AD [13][26].

Further east, the Minsk Voivodeship and the Orsha-Mogilev Dnieper corridor (Region 7, 102.278 km2) spanned the central Belarusian watershed and the Upper Dnieper, Berezina and Sozh river arteries. Its population rose from 260k in 1400AD to 430k in 1550AD and 680k in 1648AD, led by Mogilev (~25k in 1648AD, the largest city in Belarus and the principal transit market between Moscow and Poland-Lithuania), Minsk (~13k), Slutsk (~12,5k), Orsha (~9,5k), Mstislavl (~9k), the Radziwill residence at Nesvizh (~7,5k), Gomel and Bobruysk [9][49]. In the south-west, the Brest-Litovsk Voivodeship and Pinsk-Turov-Mozyr Polesia (Region 8, 62.824 km2) encompassed the Western Bug gateway at Brest-Litovsk (~15k in 1648AD, granted Magdeburg Law in 1390AD and site of the 1596AD Church Union) alongside the Pripyat river-marshland towns of Pinsk (~12k), Mozyr (~6k) and Turov, with its population tripling from 170k in 1400AD to 310k in 1550AD and 520k in 1648AD [9][13].

This growth rested on Queen Bona Sforza's and Sigismund II Augustus's Voloka Reform (*Ustava na voloki*, 1557AD), which consolidated scattered peasant *dvorishcha* into regular three-field linear villages divided into standard 21,36 ha *voloka* plots, paired with the chartering of more than 60 Magdeburg-law municipalities and over 300 noble market towns (*miasteczka*) [13][49].

Crown Ukraine and the Left Bank

Across the Ukrainian and Crimean territories (Regions 9-17, covering 607.825 km2), the total population rose from 2,51M in 1400AD to 3,16M in 1500AD, 3,71M in 1550AD, 4,66M in 1600AD and 6,1M in 1648AD [5][27]. Following the Union of Lublin (1569AD), which transferred Volhynia, Podolia-Bratslav and the Kyiv Voivodeship from Lithuania to the Polish Crown, settlement advanced steadily from the western palatinates into the Dnieper borderlands.

The Ruthenian and Belz Voivodeships of Galicia, together with Moldavian Bukovina and Hungarian Carpathian Rus (Region 9, 71.558 km2), formed the most densely settled East Slavic territory under Polish Crown rule, expanding from 820k in 1400AD to 1,18M in 1550AD and 1,75M in 1648AD (24,5 persons/km2). Its metropolitan capital Lviv (~25k in 1648AD) linked Black Sea and Levantine trade with Central Europe, flanked by Drohobych (~7,2k), Brody (~6,5k), Halych, Ternopil and Zhovkva [5][27]. To the north-east, the Volhynian Voivodeship (Region 10, 45.176 km2), seat of the great Ruthenian princely houses of Ostrogski, Zaslawski, Zbaraski, Sanguszko and Czartoryski, grew from 420k in 1400AD to 620k in 1550AD and 920k in 1648AD, anchored by Ostroh (~12k, site of the Ostroh Academy and the 1581AD Slavonic Bible), Lutsk (~11k), Kremenets (~8k), Dubno (~7,5k) and Volodymyr (~6k) [5]. Across the Dniester and Southern Bug

forest-steppe uplands, the Podolian and Bratslav Voivodeships (Region 11, 39.620 km2) rose from 350k in 1400AD to 540k in 1550AD and 910k in 1648AD, led by the royal cliff fortress of Kamianets-Podilskyi (~15k), Vinnytsia (~9k), Bratslav (~8,5k), Bar (~7,5k) and Medzhybizh (~7k) [5][27].

Further east, the Right-Bank Kyiv Voivodeship and Cossack regiments (Region 12, 56.764 km2), covering the Right-Bank Dnieper, Teterev, Irpin, Ros and Tyasmin basins, surged from 280k in 1400AD and 440k in 1550AD to 920k in 1648AD as magnate *slobody* (granting 15-30 years of tax exemption) and Registered Cossack regiments proliferated around Kyiv (~18k in 1648AD, intellectual seat of Metropolitan Petro Mohyla's Collegium), Uman (~11k), Bila Tserkva (~10,5k), Korsun (~9,5k) and Chyhyryn (~8k, Bohdan Khmelnytsky's residence and first Hetmanate capital) [5][27]. Beyond the Dnieper, in the Northern Left-Bank Hetmanate (Region 13, 55.782 km2, covering Chernihiv, Nizhyn, Starodub, Putyvl, Baturyn and Hlukhiv) and the Southern Left-Bank Hetmanate (Region 14, 52.082 km2, covering Pereyaslav, Lubny, Myrhorod, Pryluky, Poltava and Hadiach), the 1400AD populations stood at 210k and 110k respectively. After the 1618AD Truce of Deulino transferred Chernihiv-Seversk to Poland and the trans-Dnieper Wisniowiecki latifundium (*Wisniowiecczyzna*) centred on Lubny expanded from ~600 households in 1600AD to over 38.000 households (~230k inhabitants across 56 trans-Dnieper towns) by 1645AD, Region 13 reached 490k and Region 14 reached 460k in 1648AD (950k combined across the Left Bank) [5][27].

Seventeenth-Century Crises

The Time of Troubles

Following partial demographic recovery under Tsar Feodor Ivanovich and Boris Godunov (1584-1600AD) - which lifted Zamoskovny Moscow (Region 18) to 2,95M and Novgorod-Pskov (Region 19) to 1,04M in 1600AD - the Tsardom of Russia was struck by the Huaynaputina volcanic-winter famine of 1601-1603AD and the civil and international wars of the Time of Troubles (*Smuta*, 1604-1618AD) [1][50][51]. Post-*Smuta* inspection cadastres (*dozornye knigi* of 1614-1619AD) record steep contractions across central and north-western Russia between 1600AD and 1615AD [1][21].

The Moscow and Zamoskovny core (Region 18) fell by 22,7%, from 2,95M in 1600AD to 2,28M in 1615AD, amid famine, the Bolotnikov rebellion, the Tushino camp, the 1610-1612AD Polish garrison in Moscow and *Lisowczycy* light-cavalry raids. The Novgorod and Pskov Five Pyatiny (Region 19) plunged by 31,7% (and by 47% relative to 1550AD), from 1,04M in 1600AD to 710k in 1615AD, during Jacob De la Gardie's Swedish occupation of Veliky Novgorod (1611-1617AD) and Gustavus Adolphus's siege of Pskov (1615AD). Olonets Karelia and Baltic Ingria (Region 20) dropped by 25%, from 240k in 1600AD to 180k in 1615AD, mirroring Vyborg and Kexholm (Region 4, 95k to 80k). When the Treaty of Stolbovo (1617AD) ceded Baltic Ingria (*Ivangorod, Yam, Koporye and Noteburg*) alongside Kexholm County to Sweden, tens of thousands of Orthodox Karelians and Izhorians fled Swedish rule in the 1620s-1650sAD into the Bezhetskiy Verkh, Tver, Torzhok and Tikhvin *uezds* (Regions 18-19) [38][40][41].

Along the western and southern approaches, Smolensk and Seversk (Region 21) fell by 26,1% (690k to 510k) following the twenty-month Polish siege of Smolensk (1609-1611AD) and its cession at Deulino (1618AD), while Ryazan, Tula and Oryol (Region 23) declined from 1,02M to 820k and the Belgorod and Sevsk Razryads (Region 24) dipped from 410k to 380k [51][52]. Conversely, sheltered Pomorye and Vologda (Region 22, 410k to 420k), Vyatka, Perm and the Urals (Region 26, 390k to 405k) and the eight Siberian zones (Regions 31-38, 357k to 375k) escaped devastation and financed the 1611-1612AD Second Volunteer Army of Minin and Pozharsky from Nizhny Novgorod (Region 25) and Yaroslavl (Region 18) [1].

The Deluge in Lithuania and Belarus

During the Russo-Polish War of 1654-1667AD (*Potop* or *Krovavy Potop*), both Core Lithuania (Region 3, falling from 1,06M to 640k, or -39,6%) and the four Belarusian regions (Regions 5-8) suffered one of the steepest demographic contractions in early modern Europe [9][13]. Comparing the 1648-1650AD and 1667AD *podymne* hearth-tax registers across all Grand Duchy *powiaty*, J. Morzy (1965) and A. M. Karpachev and P. G. Kozlovsky (1972) showed that Belarus lost 48,8% of its total population, falling from 2,5M in 1648AD to 1,28M at the Truce of Andrusovo in 1667AD.

Within Belarus, the Polotsk and Vitebsk Voivodeships (Region 5) fell by 51,8% (560k to 270k) amid Dvina sieges, the 1657-1658AD plague and forced resettlements of skilled Belarusian townspeople (*belorustsy*) to Moscow [13]. The Minsk, Orsha and Mogilev Dnieper corridor (Region 7) dropped by 51,5% (680k to 330k) following the sack of Mstislavl (1654AD), the 1661AD Mogilev uprising and continuous campaigning along the Smolensk-Orsha-Minsk road [9]. Brest-Litovsk and Pinsk-Turov-Mozyr Polesia (Region 8) declined by 50% (520k to 260k), while the Grodno and Novogrudok Voivodeships (Region 6) contracted by 43,2% (740k to 420k) [9][13].

Although the four Belarusian regions partially recovered to 1,45M in 1678AD and 1,63M in 1700AD, the Great Northern War (1700-1721AD) - marked by Russian, Swedish, Saxon and Polish operations across Grodno, Polotsk, Lesnaya (1708AD) and Mogilev (burned in 1708AD), alongside the 1708-1711AD plague - pushed Belarus back to 1,43M in 1710AD and 1,55M in 1720AD, still 38% below its 1648AD peak [9][16][49].

The Ruin in Ukraine and Sloboda Colonisation

In Ukraine, the Khmelnytsky Uprising (1648-1657AD), the 1654AD Pereyaslav Agreement and the thirty-year multi-sided conflict known as *The Ruin* (*Ruina*, 1657-1687AD) shifted the demographic centre of gravity from Right-Bank Ukraine (Regions 10, 11 and 12) to the Left-Bank Hetmanate (Regions 13 and 14) and Sloboda Ukraine (Region 15) [5][27][53][54]. Between 1648AD and 1678AD, the Right-Bank Kyiv Voivodeship and Cossack regiments (Region 12) lost 60,9% of their population, falling from 920k in 1648AD to 520k in 1667AD, 360k in 1678AD and 320k in 1700AD after the Ottoman-Tatar Chyhyryn campaigns (1674-1678AD, which destroyed Uman and Chyhyryn), Hetman Ivan Samoilovych's 1678-1679AD *Velykyi Zhin* (forced trans-Dnieper evacuation to the Left Bank) and the 1686AD Eternal Peace buffer-zone clause, while the city of Kyiv itself remained under Russian garrison rule on the Dnieper cliff at ~15k-18k [5][53]. Simultaneously, Podolia and Bratslav (Region 11, ruled by the Ottoman Empire as the Eyalet of Kamanice in 1672-1699AD) fell by 58,2% from 910k to 380k in 1678AD before recovering to 510k by 1720AD after the 1699AD Treaty of Karlowitz; Volhynia (Region 10) dropped from 920k to 580k (recovering to 680k in 1720AD); and Galicia-Belz (Region 9) contracted from 1,75M to 1,38M (reaching 1,62M in 1720AD) [5].

Absorbing waves of Right-Bank settlers, the Northern Left-Bank Hetmanate (Region 13, covering Chernihiv, Nizhyn, Starodub and Putyvl, plus the Hetman capitals of Baturyn in 1669-1708AD and Hlukhiv in 1708-1764AD) grew continuously from 490k in 1648AD to 560k in 1667AD, 620k in 1678AD and 790k in 1720AD (+61,2%) [53]. The Southern Left-Bank Hetmanate (Region 14, comprising the Pereyaslav, Lubny, Myrhorod, Pryluky, Hadiach and Poltava regiments) rose from 460k in 1648AD to 540k in 1667AD, 610k in 1678AD and 760k in 1720AD (+65,2%), weathering the 1708-1709AD Swedish invasion and the Battle of Poltava in June 1709AD [5][53]. Further east, Sloboda Ukraine (Region 15, 60.982 km2, spanning the Kharkiv, Sumy, Okhtyrka, Ostrogozhsk and Izium regiments) was colonised by Ukrainian *cherkasy* migrants under Tsar Aleksey Mikhailovich's autonomous regimental charters (Ostrogozhsk in 1652AD, Sumy in 1652AD, Kharkiv in 1654AD, Okhtyrka in 1654AD and Izium in 1681AD), surging from 110k in 1600AD and 160k in 1648AD to 250k in 1667AD, 310k in 1678AD, 380k in 1700AD and 400k in 1720AD (+150% above 1648AD) [17][54][55].

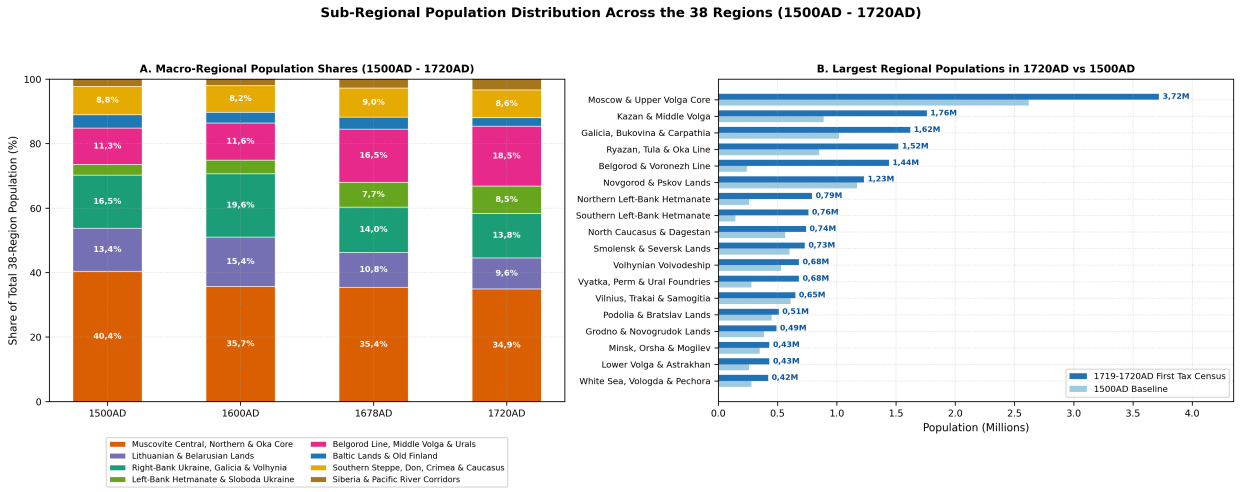


Figure 3. Regional percentage population shifts during major crisis and colonisation phases (Panel A) and Siberian population growth (1580-1720AD, Panel B).

The Southern Frontier and the Steppe

The Oka and Belgorod Lines

South of the Muscovite forest zone, the conversion of the forest-steppe chernozem from a Crimean-Nogai raiding ground (*Pole*) into the agrarian core of Russia unfolded across two frontier regions [18][52][56]. Along the Ryazan, Tula, Oryol and Great Oka Abatis Line (Region 23, 95.590 km2), settlement was shielded by the felled-forest barriers, earthworks and artillery gates of the Great Abatis Line (*Bolshaya Zasechnaya Cherta*, built in 1521-1566AD and rebuilt in 1638AD) running from Kozelsk, Belev and Odoyev through Tula and Venev to Ryazan (*Pereyaslavl-Ryazansky*), Kolomna, Oryol (1566AD) and Yelets (1592AD). This region expanded from 680k in 1400AD to 980k in 1550AD, recovered from its 1615AD *Smuta* dip (820k) and reached 1,08M in 1648AD, 1,29M in 1678AD and 1,52M in 1720AD (15,9 persons/km2), with Tula (19k in 1720AD, home to the 1632AD Vinius ironworks and the 1712AD Imperial Arms Plant) serving as Russia's leading weapons centre [7][18].

Further south, in the Belgorod and Sevsk Razryads (Region 24, 148.778 km2), the Military Chancellery (*Razryadny Prikaz*) blocked the Muravsky, Izyumsky and Kalmiussky Tatar trails across the Kursk, Belgorod, Voronezh, Lipetsk, Kozlov and Tambov chernozem *uezds* by founding Voronezh (1586AD), Livny (1586AD), Stary Oskol (1593AD), Belgorod (1596AD) and Kursk (1596AD), and in 1635-1658AD built the continuous 800 km Belgorod Fortified Line (*Belgorodskaya Cherta*) extending to Kozlov (1635AD) and Tambov (1636AD) [47][56]. Organised into the Belgorod Razryad (1658AD) and Sevsk Razryad, Region 24 grew more than seven-fold from 190k in 1400AD and 290k in 1550AD to 620k in 1648AD, 980k in 1678AD and 1,44M in 1720AD (+132,3% above 1648AD) [7][57]. Its rural society rested on *odnodvortsy* ("single-homesteaders" descended from 17th-century *deti boyarskie*, cossacks, streltsy and dragoons, exceeding 450k male souls by the 1720sAD), alongside expanding landlord serf estates and Peter I's Azov fleet shipyards at Voronezh (16k in 1720AD) and Lipetsk [18][58].

Frontier System/ Regional Polity	Active Phase	Geographic Span	Principal Urban & Garrison Hubs	Dominant Social Estates	Regional Population Trajectory
Great Oka Abatis Line (Region 23)	1521-1638AD	600 km (Kozelsk - Tula - Kolomna - Ryazan)	Tula, Ryazan, Oryol, Kolomna, Yelets, Belev	Pomeshchiki, streltsy, gunsmiths & landlords' serfs	680k (1400AD) to 1,08M (1648AD) to 1,52M (1720AD)
Belgorod & Sevsk Razryads (Region 24)	1586-1658AD	800 km (Belgorod - Voronezh - Kozlov - Tambov)	Belgorod, Voronezh, Kursk, Tambov, Kozlov, Stary Oskol	Odnodvortsy (deti boyarskie), streltsy & serfs	190k (1400AD) to 620k (1648AD) to 1,44M (1720AD)
Kazan & Simbirsk-Penza Line (Region 25)	1552-1683AD	650 km (Saransk - Karsun - Simbirsk - Penza - Syzran)	Kazan, Nizhny Novgorod, Simbirsk, Saransk, Penza, Samara	Service Tatars/ Mordvins, yasak Chuvash/Mari & serfs	780k (1400AD) to 1,18M (1648AD) to 1,76M (1720AD)
Sloboda Ukraine Regiments (Region 15)	1652-1681AD	530 km (Sumy - Okhtyrka - Kharkiv - Ostrogozhsk - Izium)	Kharkiv, Sumy, Okhtyrka, Izium, Ostrogozhsk, Valuyki	Ukrainian Cherkasy regimental cossacks & burghers	50k (1400AD) to 160k (1648AD) to 400k (1720AD)
N. & S. Left-Bank Hetmanate (Regions 13-14)	1648-1720AD	Desna, Sula, Psyol & Vorskla regiment network	Baturyn, Hlukhiv, Nizhyn, Chernihiv, Poltava, Lubny	Registered Cossacks, pospolyti peasants & burghers	320k (1400AD) to 950k (1648AD) to 1,55M (1720AD)
Bashkir Four Darugas & Yaik (Region 27)	1574-1720AD	Belaya (Agidel), Ufa & Yaik river frontier	Ufa (1574AD), Birsk (1663AD), Menzelinsk, Yaitskiy Gorodok	Votchinniki Bashkirs (4 darugas), teptyari & Yaik Cossacks	180k (1400AD) to 260k (1648AD) to 340k (1720AD)

Table 4. Defensive fortification lines, frontier razryads, Cossack regimental polities and population growth across Regions 13-15 and 23-27 (1400-1720AD).

Zaporozhia, Crimea, the Lower Volga, the Don and the North Caucasus

Beyond the settled forest-steppe lines lay five steppe, maritime and mountain regions. In the Zaporozhian Sich and Pontic Wild Fields (Region 16, 193.859 km2, rising from 60k in 1400AD to 130k in 1648AD and 180k in 1720AD), settlement clustered around the island strongholds of the Zaporozhian Cossack Host below the Dnieper rapids (Khortytsia, Tomakivka, Bazavluk, Mykytyn Rih, Chortomlyk in 1652-1709AD, Kamianka and Oleshky), the Polish river fort of Kodak (1635AD), surrounding winter fishing-pastoral homesteads (*zymivnyky*), the Ottoman Black Sea fortress of Ochakiv (*Ozu*) and the Bakhmut salt-boiling forts [27][54]. To the south, the Crimean Khanate and Ottoman Eyalet of Kefe (Region 17, 32.002 km2, growing from 210k in 1400AD to 360k in 1648AD and 370k in 1720AD) was divided after 1475AD between the direct Ottoman coastal province of Kefe (*Kaffa/Feodosia*) - holding ~65k-85k Muslim, Greek, Armenian

and Jewish inhabitants in 16th-century Ottoman *tahrir defterleri* across Kaffa, Sudak, Balaklava, Mangup, Kerch and Taman [28][59] - and the Giray Crimean Khanate (~270k-290k inhabitants), anchored by Bakhchisarai (~20k-22k), Qarasubazar (*Karasubazar*, ~11k-12,5k) and Gozleve (*Yevpatoria*, ~10k) [28].

On the Caspian-Volga frontier, the Lower Volga garrison towns, Astrakhan and the Kalmyk Khanate (Region 28, 251.397 km2) rose from 240k in 1400AD to 320k in 1648AD and 430k in 1720AD. Following Ivan IV's conquest of the Khanate of Astrakhan (1556AD), Muscovy secured the Volga-Caspian trade artery with Astrakhan (22k in 1720AD, hosting Russian, Tatar, Bukharan, Persian, Armenian and Indian merchant compounds), Tsaritsyn (1589AD) and Saratov (1590AD), while Oirat pastoralists under Kho Orluk and Ayuka Khan (1669-1724AD) established the Kalmyk Khanate (~40k-50k *kibitki* or ~200k-250k nomads) across the surrounding Caspian-Volga steppes [7][48][60]. West of the Volga bend, the Don Cossack Host and Azov (Region 29, 169.534 km2) expanded from 90k in 1400AD to 165k in 1648AD, 240k in 1700AD, 230k in 1710AD and 250k in 1720AD. Along the Lower and Middle Don, Seversky Donets, Khopyor and Medveditsa, the autonomous Don Cossack Host established over 120 fortified *gorodki* governed from Razdory and Cherkassk (capital from 1644AD, ~8,5k in 1720AD, plus Mikhaylovskiy Gorodok on the Khopyor), contesting Ottoman Azov (held by the Cossacks in 1637-1642AD and by Peter I in 1696-1711AD alongside newly built Taganrog in 1698AD) and experiencing a brief demographic dip during the Bulavin Rebellion of 1707-1708AD [47][61].

South of the steppe, the North Caucasus (Region 30, 263.210 km2, moving from 520k in 1400AD to 665k in 1648AD and 740k in 1720AD) encompassed Adyghe and Circassian societies, Greater and Lesser Kabarda (allied with Moscow from 1557AD), Vainakh (*Chechen* and *Ingush*) highland tower communities and piedmont settlements [62], the Russian Terek delta fortress of Terskiy Gorod (*Terki*, 1588AD) with the Grebensky Cossacks, and the polities of Dagestan (Derbent, the Kumyk Shamkhalate of Tarki, Endirey, Kumukh, the Avar Khanate and mountain *jamaat* unions) [63].

The Volga, the Urals and Siberia

Kazan, Perm and Bashkiria

East of the Oka and Sura rivers, the conquest of the Khanate of Kazan (1552AD) integrated the Middle Volga, Kama and Ural basins into the Russian state across three regions. In the former Khanate of Kazan, Nizhny Novgorod, the Chuvash-Mari lands and the Simbirsk-Penza Line (Region 25, 237.582 km2), the population grew from 780k in 1400AD to 940k in 1550AD, 1,18M in 1648AD and 1,76M in 1720AD. This territory combined Nizhny Novgorod (15k in 1720AD, adjacent to the Makaryev Fair), Arzamas, Kasimov and Temnikov with the former Kazan Khanate (Kazan at 30k in 1720AD, Sviyazhsk founded in 1551AD, Cheboksary in 1555AD and Samara in 1586AD) and the Simbirsk-Kama (1648-1656AD) and Penza-Syzran (1663-1683AD) fortification lines (Saransk in 1641AD, Simbirsk in 1648AD, Penza in 1663AD and Syzran in 1683AD), populated by Russian peasants alongside *yasak* and service (*sluzhilye*) Volga Tatars, Chuvash, Mordvins, Mari and Udmurts [7][64][65][66].

To the north-east, Great Perm, Vyatka, the Solikamsk-Stroganov salt lands and the early Demidov Ural foundries (Region 26, 479.789 km2) expanded from 220k in 1400AD to 330k in 1550AD, 480k in 1648AD and 680k in 1720AD, linking the *chernososhnye* grain basin of Vyatka (*Khlynov*) with Perm Velikaya (Cherdyn, Solikamsk at 9,8k in 1720AD and the Stroganov Kama-Chusovaya salt domains), the Sylva agricultural hub of Kungur (1648/1663AD) and Nikita Demidov's Petrine blast-furnace ironworks in the Middle Urals (Nevyansk in 1701AD, Kamensk in 1701AD and Alapaevsk in 1704AD) [7][8]. South of Perm, the Bashkir Four Darugas and the Yaik Cossack steppe frontier (Region 27, 484.827 km2, rising from 180k in 1400AD to 215k in 1550AD, 260k in 1648AD and 340k in 1720AD) were administered after Ufa's foundation (1574AD) through the four Bashkir *darugas* (Kazan, Nogai, Osinskaya and Siberian), where hereditary Bashkir *votchinniki* clans defended their patrimonial land rights in the uprisings of 1662-1664AD, 1681-1684AD and

1704-1711AD alongside Russian fortresses (Birsk in 1663AD and Menzelinsk), *teptyari* migrants and the Yaik Cossack Host at Yaitskiy Gorodok [7][57].

Siberian Colonisation

East of the Ural divide, Russian servicemen (*sluzhilye lyudi*), fur trappers (*promyshlenniki*) and state plough peasants (*pashennye krestyane*) advanced 6.000 km from Yermak's 1582AD capture of Qashliq (*Isker*) to Okhotsk (1647AD), the Bering Strait (1648AD), the Amur (1651-1689AD) and Kamchatka (1697AD) [6][20][67][68]. Because early modern Siberian settlement followed east-west river-portage highways (*voloki*), southern fortified ploughlands (*gosudareva desyatinnaya pashnya*), nomadic steppe borderlands and northern *yasak* fur districts, Northern Asia (12.806.014 km2 across Regions 31-38, rising from 300k in 1400AD and 338k in 1580AD to 435k in 1648AD, 545k in 1678AD and 760k in 1720AD) is divided into eight regional zones (Panel B of Figure 1, Panel B of Figure 3 and Table 5).

Along the Pomor-Mangazeya maritime and Taz-Turukhansk Arctic portage route (Region 31, 1.760.905 km2), Pomor mariners sailing *kochi* across the Kara Sea and the Yamal portage founded Obdorsk (1595AD), Mangazeya (1601AD) on the Taz River and Turukhansk (1607AD), lifting seasonal and indigenous Nenets, Enets, Nganasan and Selkup numbers from 16k in 1400AD and 19k in 1580AD to a peak of 28k in 1615AD and 26k in 1648AD, before Tsar Mikhail Romanov's 1619AD decree closing the Mangazeya Sea Route, sable depletion and the 1642AD Mangazeya fire relocated administration to Novaya Mangazeya (*Turukhansk*, 1672AD) and reduced Region 31 to 18k by 1720AD [6][67][69]. South of the Arctic circle, the Tobolsk-Irtysh-Middle Ob river portage highway (Region 32, 871.344 km2, growing from 42k in 1400AD and 49k in 1580AD to 88k in 1648AD, 118k in 1678AD and 158k in 1720AD) was controlled via the Babylka Ural Portage (1598AD) at Verkhoturye (1598AD), Turinsk (1600AD), Pelym (1593AD), Tyumen (1586AD) and the Siberian capital Tobolsk (1587AD, 15k in 1720AD), extending down the Lower Irtysh (Berezov in 1593AD) and Middle Ob (Surgut in 1594AD, Narym in 1596AD and Ketsky Ostrog in 1602AD) across Khanty, Mansi and Siberian Tatar *yasak* volosts [7][70][71][72].

The southern Tobol-Ishim-Tara-Tomsk-Kuznetsk fortified agricultural frontier (Region 33, 359.192 km2, rising from 38k in 1400AD and 47k in 1580AD to 86k in 1648AD, 145k in 1678AD and 254k in 1720AD) stretched along the forest-steppe chernozem from the Iset-Tobol *slobody* (Shadrinsk in 1662AD and Kurgan in 1679AD), Tara (1594AD) and Omsk (1716AD) to Tomsk (1604AD, 8,5k in 1720AD) and Kuznetsk (1618AD). Region 33 became the main demographic engine of Siberia, expanding more than five-fold between 1580AD and 1720AD and housing over 60% of Northern Asia's Russian plough peasantry [20][73][74]. Immediately to its south, the nomadic Baraba-Altai-Minusinsk-Tuva steppe borderlands (Region 34, 821.185 km2, moving from 74k in 1400AD and 81k in 1580AD to 84k in 1648AD, 80k in 1678AD and 100k in 1720AD) remained a contested zone of dual tribute (*dvoedanstvo*) inhabited by Baraba Tatars, Teleuts, Shors, Tuvans and the four Yenisei Kirghiz *ulusy* of Minusinsk until the 1703AD Dzungar relocation of the Kirghiz elite enabled Russian construction of Abakansky Ostrog (1707AD) and Biysky Ostrog (1709AD) [6][73].

In Central and Eastern Siberia, the Ket-Yeniseysk-Angara-Ilimsk-Irkutsk river portage spine (Region 35, 1.931.504 km2, growing from 28k in 1400AD and 32k in 1580AD to 44k in 1648AD, 66k in 1678AD and 98k in 1720AD) linked the Makovskiy Ket-Yenisei portage (1618AD) through Yeniseysk (1619AD, 8k in 1720AD) and Krasnoyarsk (1628AD) up the Angara to Ilimsk (1630AD, on the Lena Portage) and Irkutsk (1661AD, 6,5k in 1720AD) among Western Buryats, Kets and Evenks [6][7][67]. Beyond Lake Baikal, Transbaikal Dauria and the Upper Amur frontier (Region 36, 1.753.934 km2, rising from 34k in 1400AD and 38k in 1580AD to 40k in 1648AD, 48k in 1678AD and 62k in 1720AD) was anchored by Nerchinsk (1658AD), Selenginsk (1665AD) and Udinsk (1666AD) among Khori-Selenga Buryats, Evenks and Daurs, after Qing sieges of Albazin (1651-1689AD) and the Treaty of Nerchinsk (1689AD) redirected Russian settlement into Nerchinsk silver-lead mining (1704AD) and Kiakhta-Selenga caravan trade [6][75][76].

To the north-east, the central Lena-Vilyuy-Aldan basin (Region 37, 1.373.412 km2, moving from 38k in 1400AD to 42k in 1580AD and 1648AD, 44k in 1678AD and 47k in 1720AD) centred on Yakutsk (1632AD), Vilyuysk (1634AD) and Olekminsk (1635AD) in the thermokarst *alaas* meadows of the Yakut (Sakha) cattle- and horse-breeding pastoralists (~28,5k-35k Sakha across 35 *yasak* volosts plus Evenks and ~4k-5k Russians by 1720AD) [6][77]. Finally, across the Northeast Arctic and Pacific *yasak* frontier (Region 38, 3.924.538 km2, declining from 30k in 1400AD and 1580AD to 25k in 1648AD and 1678AD and 23k in 1720AD), spanning Okhotsk (1647AD), Anadyrsk (1649AD) and Nizhnekamchatsk (1697/1703AD), virgin-soil smallpox epidemics (1651-1653AD, 1669AD and 1691-1694AD) and frontier conflict reduced Yukaghir, Itelmen, Koryak, Even and Chukchi populations from 30k to 23k [6][76][78].

Siberian Zone (Reg. 31-38)	Area	Principal Ostrogi, Portages & Indigenous Ulusy	1580AD	1615AD	1648AD	1678AD	1720AD
31. Pomor-Mangazeya Maritime & Taz Portage	1.760.905 km2	Obdorsk (1595AD), Mangazeya (1601AD; closed 1619AD), Turukhansk (1607/1672AD)	19k	28k	26k	19k	18k
32. Tobolsk-Irtysh-Middle Ob Portage Highway	871.344 km2	Tyumen (1586AD), Tobolsk (1587AD), Berezov (1593AD), Surgut (1594AD), Verkhoturye (1598AD)	49k	64k	88k	118k	158k
33. S. Tobol-Tara-Tomsk-Kuznetsk Ploughland	359.192 km2	Tara (1594AD), Tomsk (1604AD), Kuznetsk (1618AD), Shadrinsk (1662AD), Omsk (1716AD)	47k	60k	86k	145k	254k
34. Nomadic Baraba-Altai-Minusinsk-Tuva Steppe	821.185 km2	Baraba & Teleut ulusy, Hongorai Kirghiz; Abakansky (1707AD), Biysky (1709AD)	81k	82k	84k	80k	100k
35. Ket-Yeniseysk-Angara-Ilimsk-Irkutsk Spine	1.931.504 km2	Makovskiy (1618AD), Yeniseysk (1619AD), Krasnoyarsk (1628AD), Ilimsk (1630AD), Irkutsk (1661AD)	32k	34k	44k	66k	98k

(Continued from previous page)

Siberian Zone (Reg. 31-38)	Area	Principal Ostrogi, Portages & Indigenous Ulusy	1580AD	1615AD	1648AD	1678AD	1720AD
36. Transbaikal Dauria & Upper Amur Frontier	1.753.934 km2	Albazin (1651-1689AD), Nerchinsk (1658AD), Selenginsk (1665AD), Udinsk (1666AD)	38k	39k	40k	48k	62k
37. Central Lena-Vilyuy-Aldan (Yakutsk Alaas)	1.373.412 km2	Yakutsk (1632AD), Vilyuysk (1634AD), Olekminsk (1635AD), Amginskaya Sloboda (1652AD)	42k	41k	42k	44k	47k
38. NE Arctic & Pacific Yasak (Kolyma-Kamchatka)	3.924.538 km2	Verkhoyansk (1638AD), Srednekolymsk (1643AD), Okhotsk (1647AD), Anadyrsk (1649AD), Nizhnekamchatsk (1697AD)	30k	27k	25k	25k	23k
Total Siberia & Far East (Regions 31-38)	12.806.014 km2	Over 50 ostrogi, slobody, portage stations and goroda across Northern Asia	338k (2k RU)	375k (25k RU)	435k (85k RU)	545k (165k RU)	760k (365k RU)

Table 5. Land areas, principal towns and benchmark population series across the eight Siberian river, steppe and frontier zones (1580-1720AD).

Petrine Reforms and the Urban Hierarchy

The reign of Peter I (1682/1689-1725AD) and the Great Northern War (1700-1721AD) reshaped both the administrative structure and the urban hierarchy of the realm [7][8][24]. After capturing Swedish Noteburg (Shlisselburg) and Nyenskans on the Neva in Region 20, Peter I founded Saint Petersburg (May 1703AD) and the naval fortress of Kronstadt (1704AD), reinforced in 1710AD by the conquest of Vyborg and Kexholm (Region 4) alongside Riga, Pernau, Arensburg and Reval (Region 1). Driven by annual conscription of 20k-40k *rabotnye lyudi* labourers and the mandatory transfer of noble families, foreign-trade merchants and central colleges (1712-1718AD), Saint Petersburg expanded from ~5k in

1703AD to ~18k in 1710AD and 40k in 1720AD, lifting Region 20 from 270k in 1700AD to 350k in 1720AD and making the new Baltic capital the second largest city in the Empire behind Moscow (Region 18, 160k) [7][24].

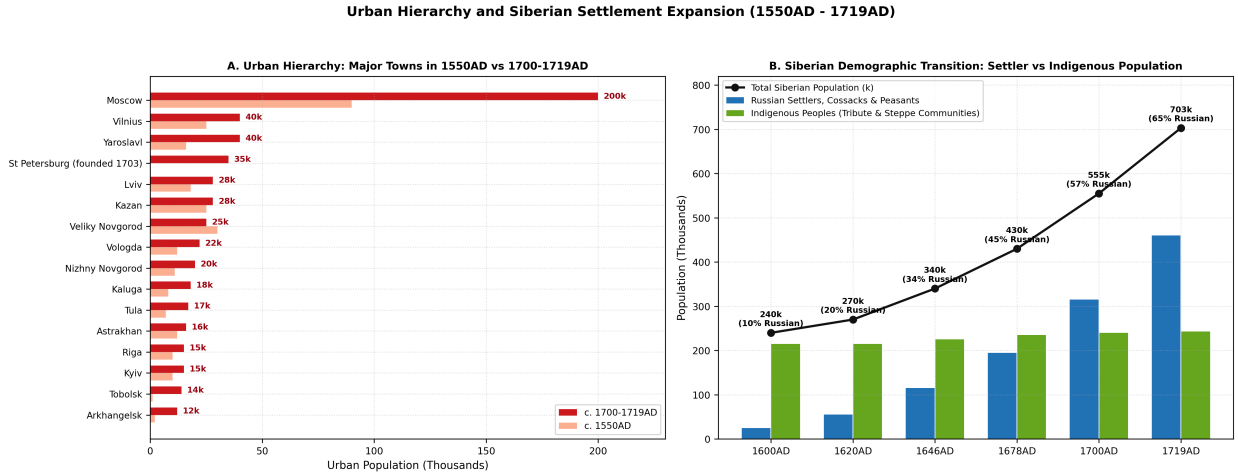


Figure 4. Largest urban centres in 1500AD versus 1648AD (Panel A) and leading cities at the First Poll-Tax Revision of 1720AD (Panel B).

Following the 1708-1710AD guberniya reform (creating Moscow, Ingermanland/Saint Petersburg, Kiev, Smolensk, Arkhangelogorod, Kazan, Azov/Voronezh and Siberia, joined after 1710-1713AD by Riga/Livland and Reval/Estland) and the 1710AD household audit - which exposed wartime recruitment losses, plague mortality in the western provinces and widespread household concealment (*utayka dvorov*, reflected in the aggregate dip from 22,06M in 1700AD to 21,55M in 1710AD) - the Senate enacted the First Poll-Tax Revision (1718-1721AD) [7][8][22]. By 1720AD, the urban hierarchy across all thirty-eight regions (Figure 4) fell into three main regional groups.

Across Russia and Siberia (Regions 18-38, 14,37M total), behind Moscow (Region 18, 160k) and Saint Petersburg (Region 20, 40k), the leading urban centres were Kazan (Region 25, 30k), Yaroslavl (Region 18, 26k), Astrakhan (Region 28, 22k), Kaluga (Region 18, 20k), Tula (Region 23, 19k), Voronezh (Region 24, 16k), Nizhny Novgorod (Region 25, 15k), Tobolsk (Region 32, 15k), Pskov (Region 19, 15k), Veliky Novgorod (Region 19, 14,5k), Kostroma (Region 18, 14,5k), Vologda (Region 22, 14k), Smolensk (Region 21, 14k), Kursk (Region 24, 12,5k), Kolomna (Region 23, 12,5k), Belgorod (Region 24, 11k) and Oryol (Region 23, 11k) [7][24].

In Ukraine and Crimea (Regions 9-17, 5,65M total), Lviv (Region 9, 25k), Bakhchisarai (Region 17, 20k) and Kaffa (Region 17, 19k) led the western and southern rims, while Kyiv (Region 12, 18k), Nizhyn (Region 13, 14k), Chernihiv (Region 13, 13,5k), Hlukhiv (Region 13, 12,5k), Poltava (Region 14, 12k), Starodub (Region 13, 11,5k), Pereyaslav (Region 14, 11,5k), Kamianets-Podilskyi (Region 11, 11,5k) and Kharkiv (Region 15, 11k) anchored the Hetmanate, Sloboda Ukraine and Podolia [5][28][53].

In the Baltic provinces, Courland, Core Lithuania, Belarus and Old Finland (Regions 1-8, 2,81M total), which were still recovering from the 1708-1711AD plague and Great Northern War sieges, Mogilev (Region 7, 18k) and Vilnius (Region 3, 13,5k, down from 28k in 1648AD) led the Lithuanian-Belarusian interior alongside Slutsk (Region 7, 9,5k), Polotsk (Region 5, 9k), Vitebsk (Region 5, 8,5k), Grodno (Region 6, 8,5k), Minsk (Region 7, 8k), Brest-Litovsk (Region 8, 8k) and Kaunas (Region 3, 6,5k), while Riga (Region 1, 9,5k), Reval/Tallinn (Region 1, 6,8k), Mitau/Jelgava (Region 2, 4,2k), Narva (Region 1, 4,2k), Vyborg (Region 4, 3,5k), Libau (Region 2, 3,2k) and Dorpat/Tartu (Region 1, 2,8k) anchored the Baltic littoral [3][9][10][35].

Conclusion

At the Treaty of Nystad (30 August/10 September 1721AD), Peter I assumed the title of Emperor (*Imperator Vserossiyskiy*), closing the Tsardom of Russia period (1400-1721AD, 38 regions, 18.119.929 km2 and 448.088 grid cells, with 22,83M inhabitants in 1720AD) and linking directly into the Early Imperial Russia series (1721-1815AD, 40 regions, 18.496.636 km2 and 457.996 grid cells, with 23,89M inhabitants in 1721AD) [8][57]. Across the newly annexed Nystad and Polish-Lithuanian border provinces, regional populations align directly between the two series: Region 1 (300k in 1720AD) divides into Estland Governorate (105k in 1721AD) and Livland Governorate (195k in 1721AD); Region 2 (200k) corresponds to Courland (200k); Region 3 (650k) aligns with Vilna and Samogitia (655k); and Region 4 (110k) continues as Old Finland/Vyborg Governorate (110k). The net difference of +1,06M (+376.707 km2 and +9.908 grid cells) between the 1720AD Tsardom terminal total (22,83M) and the 1721AD Early Imperial baseline (23,89M) reflects two components: first, a +590k territorial addition incorporating the Grand Duchy of Finland (annexed in 1809AD at Fredrikshamn, 333.899 km2, 390k in 1721AD) and Bessarabia/Budjak (annexed in 1812AD at Bucharest, 47.077 km2, 200k in 1721AD) so that the 1721-1815AD series maintains constant boundaries; and second, an increase of +468k (+2,05%) across the continuous 38-region core from post-Nystad recovery and the military revision *popolnenie* audits of 1721-1724AD [7][8].

Sources

1. Vodarsky, Yaroslav Evgenevich. *Naselenie Rossii Za 400 Let (XVI - Nachalo XX Vv.) [the Population of Russia over 400 Years, 16th to Early 20th Centuries]*. Prosveshchenie, 1973, https://search.worldcat.org/search?q=Population+Russia+over+400+Years+16th+Early+20th+Vodarsky+1973.
2. Kopanev, Aleksandr Ilich. “Naselenie Russkogo Gosudarstva v XVI v. [The Population of the Russian State in the 16th Century].” *Istoricheskie Zapiski*, vol. 64, 1959, pp. 233-54, https://cyberleninka.ru/article/n/naselenie-russkogo-gosudarstva-v-xvi-v.
3. Palli, Heldur. *Eesti Rahvastiku Ajalugu Kuni 1712. Aastani [Population History of Estonia up to 1712]*. Teaduste Akadeemia Kirjastus, 1996, https://search.worldcat.org/search?q=Population+History+Estonia+up+1712+Palli+1996.
4. Ochmanski, Jerzy. *Historia Litwy [History and Demographic Reconstruction of the Grand Duchy of Lithuania]*. Ossolineum, 1982, https://search.worldcat.org/search?q=History+Demographic+Reconstruction+Grand+Duchy+Lithuania+Ochmanski+1982.
5. Yakovenko, Nataliya Mykolaivna. *Narys Istorii Serednovichnoi Ta Rannomodernoi Ukrainy [Outline History of Medieval and Early Modern Ukraine]*. Krytyka, 2006, https://search.worldcat.org/search?q=Outline+History+Medieval+Early+Modern+Ukraine+Yakovenko+2006.
6. Dolgikh, Boris Osipovich. *Rodovoy i Plemennoy Sostav Narodov Sibiri v XVII Veke [Clan and Tribal Composition of the Peoples of Siberia in the 17th Century]*. Izdatelstvo Akademii Nauk SSSR, 1960, https://search.worldcat.org/search?q=Clan+Tribal+Composition+Peoples+Siberia+17th+Century+Dolgikh+1960.
7. Vodarsky, Yaroslav Evgenevich. *Naselenie Rossii v Kontse XVII - Nachale XVIII Veka (Chislennost, Soslovno-Klassovyy Sostav, Razmeshchenie) [the Population of Russia in the Late 17th and Early 18th Centuries: Size, Social Composition and Distribution]*. Nauka, 1977, https://doi.org/10.2307/1853555.
8. Kabuzan, Vladimir Maksimovich. *Narodonaselenie Rossii v XVIII - Pervoy Polovine XIX v. (Po Materialam Reviziy) [Population of Russia in the 18th and First Half of the 19th Century According to the Revisions]*. Izdatelstvo Akademii Nauk SSSR, 1963, https://search.worldcat.org/search?q=Population+Russia+18th+First+Half+19th+Century+According+Kabuzan+1963.
9. Karpachev, Avram Moiseevich, and Pavel Grigorevich Kozlovsky. *Dinamika Chislennosti Naseleniya*

Belorussii Vo Vtoroy Polovine XVII - XVIII Vv. [Dynamics of the Population Size of Belarus in the Second Half of the 17th and 18th Centuries]. Nauka i Tekhnika, 1972, pp. 81-94, https://search.worldcat.org/search?q=Dynamics+Population+Size+Belarus+Second+Half+17th+18th+Karpachev+1972.

10. Dunsdorfs, Edgars. *Latvijas Vesture 1600-1710 [History and Historical Demography of Latvia, Livonia, Courland and Latgale, 1600-1710]*. Daugava, 1962, https://search.worldcat.org/search?q=History+Historical+Demography+Latvia+Livonia+Courland+Latgale+1600+Dunsdorfs+1962.
11. Saloheimo, Veijo. *Pohjois-Karjalan Historia II: 1617-1721 [History and Population Resettlement of Kexholm County and Northern Karelia, 1617-1721]*. Karjalaisen Kulttuurin Edistamissaatio, 1976, https://search.worldcat.org/search?q=History+Population+Resettlement+Kexholm+County+Northern+Karelia+1617+Saloheimo+1976.
12. Shapiro, Aleksandr Lvovich. *Agrarnaya Istoriya Severo-Zapada Rossii XVI Veka: Novgorodskie Pyatiny [Agrarian History of North-Western Russia in the 16th Century: The Novgorodian Pyatiny]*. Nauka, 1974, https://search.worldcat.org/search?q=Agrarian+History+North+Western+Russia+16th+Century+Novgorodian+Shapiro+1974.
13. Morzy, Jozef. *Kryzys Demograficzny Na Litwie i Bialorusi w II Polowie XVII Wieku [the Demographic Crisis in Lithuania and Belarus in the Second Half of the 17th Century]*. Wydawnictwo Naukowe Uniwersytetu im. Adama Mickiewicza, 1965, https://search.worldcat.org/search?q=Demographic+Crisis+Lithuania+Belarus+Second+Half+17th+Century+Morzy+1965.
14. Seppel, Marten. "The Assistance of Manors to Peasants in Cases of Subsistence Crisis in Seventeenth-Century Estonia and Livonia." *Acta Historica Tallinnensia*, vol. 8, 2004, pp. 3-19, https://doi.org/10.3176/hist.2004.1.01.
15. Malve, Martin. "First Glimpse of Infectious Diseases in Medieval and Early Modern Estonia Based on Osteoarchaeological Material." *Estonian Journal of Archaeology*, vol. 27, no. 3S, 2023, pp. 165-84, https://doi.org/10.3176/arch.2023.3S.08.
16. Majewska, Gabriela. "The Polish-Lithuanian Commonwealth and Its Inhabitants During the Great Northern War (1700-1721)." *Studia Maritima*, vol. 31, 2018, pp. 125-46, https://doi.org/10.18276/sm.2018.31-07.
17. Bagaley, Dmitriy Ivanovich. *Ocherki Iz Istorii Kolonizatsii i Byta Stepnoy Okrainy Moskovskogo Gosudarstva [Essays on the History of Colonisation and Life on the Steppe Frontier of the Muscovite State]*. Universitetskaya Tipografiya, 1887, https://search.worldcat.org/search?q=Essays+History+Colonisation+Life+Steppe+Frontier+Muscovite+State+Bagaley+1887.
18. Stevens, Carol Belkin. *Soldiers on the Steppe: Army Reform and Social Change in Early Modern Russia*. Northern Illinois University Press, 1995, https://search.worldcat.org/search?q=Soldiers+Steppe+Army+Reform+Social+Change+Early+Modern+Stevens+1995.
19. Moon, David. *The Russian Peasantry 1600-1930: The World the Peasants Made*. Longman, 1999, https://doi.org/10.1093/ehr/116.465.229.
20. Pokrovsky, Nikolay Nikolaevich. *Tomsk 1648-1649 Gg.: Voevodskaya Vlast i Zemskie Miry [Tomsk in 1648-1649: Voivode Authority and Local Communities]*. Nauka, 1984, https://search.worldcat.org/search?q=Tomsk+1648+1649+Voivode+Authority+Local+Communities+Pokrovsky+1984.
21. Avdeev, Aleksandr Aleksandrovich. "Dorevizskiy Uchet Naseleniya v Rossii (XIII-XVII Veka): Istoriografiya i Istochniki [Pre-Revision Population Registration in Russia (13th-17th Centuries): Historiography and Sources]." *Istoricheskaya Demografiya*, vol. 1, 2025, pp. 12-38, https://doi.org/10.17323/demreview.v12i1.26573.
22. Ulyanova, Galina Nikolaevna. "Revizskie Skazki Kak Istochnik v Istoriografii 1970-Kh - 2010-Kh Gg. [Revision Tales as a Source in the Historiography of the 1970s-2010s]." *Istoricheskiy Zhurnal: Nauchnye Issledovaniya*, no. 4, 2016, pp. 412-25,

https://cyberleninka.ru/article/n/revizskie-skazki-kak-istochnik-v-istoriografii-1970-h-2010-h-godov.

23. Shapiro, Aleksandr Lvovich. *Agrarnaya Istoriya Severo-Zapada Rossii: Vtoraya Polovina XV - Nachalo XVI v. [Agrarian History of North-Western Russia: Second Half of the 15th to Early 16th Century]*. Nauka, 1971, https://search.worldcat.org/search?q=Agrarian+History+North+Western+Russia+Second+Half+15th+Shapiro+1971.
24. Mironov, Boris Nikolaevich. *Novaya Istoricheskaya Demografiya Imperskoy Rossii [New Historical Demography of Imperial Russia]*. Dmitriy Bulanin, 2007, https://cyberleninka.ru/article/n/novaya-istoricheskaya-demografiya-imperskoy-rossii-analiticheskiy-obzor-sovremennoy-literatury-1.
25. Vahtre, Sulev. *Eestimaa Talurahvas Hingeloenduste Andmeil (1782-1858) [the Peasantry of Estland According to the Soul Revisions and Earlier Land Cadastres]*. Eesti Raamat, 1973, https://search.worldcat.org/search?q=Peasantry+Estland+According+Soul+Revisions+Earlier+Land+Cadastres+Vahtre+1973.
26. Lowmianski, Henryk. *Studia Nad Dziejami Wielkiego Ksiestwa Litewskiego [Studies on the History and Demography of the Grand Duchy of Lithuania]*. Wydawnictwo Naukowe UAM, 1983, https://search.worldcat.org/search?q=Studies+History+Demography+Grand+Duchy+Lithuania+Lowmianski+1983.
27. Krypiakevych, Ivan Petrovych. *Bohdan Khmelnytskyi [Bohdan Khmelnytsky and the Social-Demographic Transformation of 17th-Century Ukraine]*. Vydavnytstvo AN URSR, 1954, https://search.worldcat.org/search?q=Bohdan+Khmelnytsky+Social+Demographic+Transformation+17th+Century+Ukraine+Krypiakevych+1954.
28. Ozturk, Yucel. "An Eastern European Capital City: Kefe and the Demography of the Crimean Khanate and Ottoman Eyâlet (1475-1700)." *Karadeniz Arastirmalari*, vol. 16, no. 62, 2019, pp. 185-212, https://doi.org/10.22378/kio.2019.1.45-70.
29. Bernotas, Rivo. "Estonian Small Towns in the Middle Ages: Archaeology and the History of Urban Defense." *Ajalooline Ajakiri / The Estonian Historical Journal*, no. 3, 2013, pp. 265-97, https://doi.org/10.12697/AA.2013.3.01.
30. Karjaharm, Toomas. *Ida Ja Laane Vahel: Eesti-Vene Suhted [Between East and West: Estonian-Russian Relations and Baltic Demography]*. Eesti Entsuklopeediakirjastus, 1998, https://search.worldcat.org/search?q=Between+East+West+Estonian+Russian+Relations+Baltic+Demography+Karjaharm+1998.
31. Plath, Ulrike, and Kaarel Vanamolder. "Clashing Weatherlands in 17th-Century Livonia: Climate, Subsistence and the Great Famine of 1695-1697." *Methis: Studia Humaniora Estonica*, vol. 24, no. 30, 2022, pp. 38-64, https://doi.org/10.7592/methis.v24i30.22101.
32. Rachuba, Andrzej. "Livonian and Courlander Nobility in Samogitia in the 17th and 18th Century." *Klio: Czasopismo Poswiecone Dziejom Polski i Powszechnym*, vol. 55, no. 3, 2020, pp. 85-110, https://doi.org/10.12775/klio.2020.039.
33. Petersone-Gordina, Elina, and Guntis Gerhards. "Dental Disease and Urban Mortality in a 17th-18th Century Community in Jelgava (Mitau), Duchy of Courland." *Papers on Anthropology*, vol. 20, 2011, pp. 311-26, https://scholar.google.com/scholar?q=Dental+Disease+Urban+Mortality+17th+18th+Century+Community+Petersone+2011.
34. Janicki, Arkadiusz. "Russian Expansion in the Baltic in the 18th Century: Demographic Losses of the Great Northern War and Plague." *Studia Historica Gedanensia*, vol. 13, 2022, pp. 101-23, https://doi.org/10.4467/23916001hg.22.006.17426.
35. Kiaupa, Zigmantas, et al. *The History of Lithuania Before 1795*. Lithuanian Institute of History, 2000, https://search.worldcat.org/search?q=History+Lithuania+Before+1795+Kiaupa+2000.
36. Vasiliauskas, Ernestas. "The Development of Towns of the Siauliai Crown Estate in Samogitia from the 16th to the 18th Century." *Archaeologia Baltica*, vol. 18, 2012, pp. 186-205, https://doi.org/10.15181/ab.v16i0.35.
37. Zvirblys, Atas. "Early Modern Urban Material Culture and Trade Connections in 17th-Century Vilnius."

Estonian Journal of Archaeology, vol. 25, no. 2, 2021, pp. 93-112, https://doi.org/10.3176/arch.2021.2.01.

38. Katajala, Kimmo. “Liniya, Zona Ili Resheto? Kontseptualizatsiya Rossiysko-Shvedskoy Granitsy 1617 g. [Line, Zone or Sieve? Conceptualising the Russo-Swedish Border of Stolbovo 1617 in Kexholm and Karelia].” *Nordic and Baltic Studies Review*, no. 2, 2017, pp. 42-65, https://doi.org/10.15393/j103.art.2017.755.
39. Selin, Adrian Aleksandrovich. “Izgnannye Iz Livonii: Ob Ustroystve Pomeshchikov Starykh i Novykh v Novgorodskoy Zemle v Kontse XVI v. [Exiled from Livonia: The Resettlement of Service Landholders in the Novgorod Land in the Late 16th Century].” *Novgorodskiy Istoricheskiy Sbornik*, vol. 19, 2020, pp. 175-98, https://doi.org/10.21638/spbu19.2020.202.
40. Fishman, Olga Mikhaylovna. “The Ethnography and Demography of Karelians Outside Karelia: Tver, Tikhvin and Valdai Resettlement.” *Nordic and Baltic Studies Review*, no. 5, 2020, pp. 218-42, https://doi.org/10.15393/j103.art.2020.1648.
41. Stepanova, Yuliya Vladimirovna. “Pervye Karelskie Vykhodtsy v Pogostakh Tverskoy Poloviny Bezhetskoy Pyatiny Po Dannym Pistsovykh Opisaniy Vtoroy Chetverti XVII v. [The First Karelian Migrants in the Tver Half of Bezhetskaya Pyatina According to Cadastres of the Second Quarter of the 17th Century].” *Ezhegodnik Finno-Ugorskikh Issledovaniy*, vol. 18, no. 1, 2024, pp. 102-16, https://doi.org/10.35634/2224-9443-2024-18-3-435-445.
42. Tikhomirov, Mikhail Nikolaevich. *Rossiya v XVI Stoletii [Russia in the 16th Century]*. Izdatelstvo Akademii Nauk SSSR, 1962, https://search.worldcat.org/search?q=Russia+16th+Century+Tikhomirov+1962.
43. Tsybakov, Denis Leonidovich. “Problemy Opredeleniya Vladeteley Territorii Verkhney Oki Na Rubezhe XV-XVI Vv. [Problems of Identifying Rulers and Principalities of the Upper Oka Territory at the Turn of the 15th-16th Centuries].” *Istoriya: Fakty i Simvoly*, no. 3, 2022, pp. 44-56, https://doi.org/10.52070/2500-347X_2022_4_849_110.
44. Postnikov, Arseniy Borisovich. “Posadskie Lyudi Pskova v Istoriografii [the Posad Population of Pskov in Historiography].” *Pskov: Nauchno-Prakticheskiy, Istoriko-Kraevedcheskiy Zhurnal*, no. 54, 2021, pp. 28-41, https://cyberleninka.ru/article/n/posadskie-lyudi-pskova-v-istoriografii.
45. Bogdanov, Vladimir Olegovich. “Sukhoputnye Dorogi Uezda Rzhevy Volodimerovoy v XV-XVII Vv. [Overland Roads and Settlement Network of Rzheva Volodimerova Uezd in the 15th-17th Centuries].” *Istoricheskaya Geografiya*, vol. 7, 2025, pp. 104-36, https://doi.org/10.24888/2410-4205-2025-42-1-8-21.
46. Hellie, Richard. *Enserfment and Military Change in Muscovy*. University of Chicago Press, 1971, https://doi.org/10.7208/chicago/9780226326467.001.0001.
47. Davies, Brian L. *Warfare, State and Society on the Black Sea Steppe, 1500-1700*. Routledge, 2007, https://doi.org/10.4324/9780203961766.
48. Moiseev, Maksim Vladimirovich. “Stepnye Voyny Ot Ugry Do Nogayskogo Pogroma Kryma (1480-1523 Gg.) [Steppe Wars from the Stand on the Ugra to the Nogai Devastation of Crimea, 1480-1523].” *Zolotoordynskaya Tsivilizatsiya*, no. 8, 2015, pp. 252-74, https://cyberleninka.ru/article/n/stepnye-voyny-ot-ugry-do-nogayskogo-pogroma-kryma-1480-1522-gg.
49. Sharukho, Igor Nikolaevich. “Mestechki Belarusi: Iz Proshlogo v Budushchee [the Miasteczka of Belarus: Urban-Agrarian Settlement Evolution from the 15th to the 20th Century].” *Pskovskiy Regionologicheskiy Zhurnal*, no. 10, 2010, pp. 115-24, https://cyberleninka.ru/article/n/mestechki-belarusi-iz-proshlogo-v-buduschee.
50. Hellie, Richard. *The Economy and Material Culture of Russia, 1600-1725*. University of Chicago Press, 1999, https://doi.org/10.2307/2652607.
51. Zorin, Aleksandr Vasilevich. “Aleksandr Yuzef Lisovskiy: Geroy Smutnogo Vremeni [Aleksander Jozef Lisowski and the Raids of the Time of Troubles in Seversk and Central Russia].” *Istoriya*

Voenno-Politicheskikh Konfliktov, vol. 4, 2012, pp. 1-78, https://cyberleninka.ru/article/n/aleksandr-yuzef-lisovskiy-geroy-smutnogo-vremeni.

52. Papkov, Andrey Igorevich. "Osvoenie Rossiyskim Gosudarstvom Yuzhnykh Okrain v XVI-XVII Vekakh [Colonisation of the Southern Borderlands by the Russian State in the 16th-17th Centuries]." *Nauchnye Vedomosti Belgorodskogo Gosudarstvennogo Universiteta: Istoriya, Politologiya*, vol. 25, no. 1, 2013, pp. 82-90, https://cyberleninka.ru/article/n/osvoenie-rossiyskim-gosudarstvom-yuzhnyh-okrain-v-xvi-veke.
53. Irtenina, Natalya Valerevna. "Evolyutsii Malorossiyskogo Kazachestva v XVII-XVIII Vv. [Social and Demographic Evolution of the Left-Bank Ukrainian Cossack Hetmanate in the 17th-18th Centuries]." *Voprosy Istorii*, no. 6, 2022, pp. 92-107, https://cyberleninka.ru/article/n/evolyutsii-malorossiyskogo-kazachestva-v-xviii-nachale-xx-v-iz-istorii-odnogo-roda.
54. Papkov, Andrey Igorevich. "Rasselenie Cherkas Na Yuzhnoy Okraine Rossii v Kontse XVI - Nachale XVIII Veka [the Settlement of Ukrainian Cherkasy on the Southern Frontier of Russia from the Late 16th to the Early 18th Century]." *Rusin*, no. 3, 2013, pp. 55-71, https://cyberleninka.ru/article/n/rasselenie-cherkas-na-yuzhnoy-okraine-rossii-v-kontse-xvi-nachale-xvii-v.
55. Krasnova, Irina Vladimirovna. "Istoki Formirovaniya Slobozhanshchiny Kak Chasti Geokulturnogo Prostranstva Rossii i Ukrainy [Origins of the Formation of Sloboda Ukraine as a Frontier Demographic Space]." *Via in Tempore: Istoriya, Politologiya*, vol. 47, no. 2, 2020, pp. 358-67, https://doi.org/10.34685/HI.2020.26.95.011.
56. Zhigalov, Vladimir Mikhaylovich. "Formirovanie i Periodizatsiya Voenno-Administrativnoy Sistemy Belgorodskogo Razryada v Kontse XVI - XVII Vv. [Formation and Periodisation of the Military-Administrative System of the Belgorod Razryad in the Late 16th-17th Centuries]." *Izvestiya Voronezhskogo Gosudarstvennogo Pedagogicheskogo Universiteta*, no. 1, 2024, pp. 64-73, https://doi.org/10.52575/2687-0967-2024-51-1-102-115.
57. Kabuzan, Vladimir Maksimovich. *Izmeneniya v Razmeshchenii Naseleniya Rossii v XVIII - Pervoy Polovine XIX v. [Changes in the Distribution of the Population of Russia in the 18th and First Half of the 19th Century]*. Nauka, 1971, https://search.worldcat.org/search?q=Changes+Distribution+Population+Russia+18th+First+Half+19th+Kabuzan+1971.
58. Astafev, Dmitriy Aleksandrovich. "Osobennosti Izucheniya Genealogii Odnodvortsev Yuzhnoy Rossii XVII-XVIII Vv. [Specifics of Studying the Genealogy and Cadastral Registration of Odnodvortsy in Southern Russia in the 17th-18th Centuries]." *Istoricheskiy Vestnik*, vol. 39, 2022, pp. 118-35, https://doi.org/10.18287/2542-0445-2022-28-4-24-32.
59. Kurnikova, Oksana Mikhaylovna. "Krepost Taman v Strukture Osmanskoy Provintsii Kefe v XVI-XVIII Vekakh [Fortress Taman in the Structure of the Ottoman Province of Kefe in the 16th-18th Centuries]." *Materialy Po Arkheologii, Istorii i Etnografii Tavrii*, vol. 31, 2026, pp. 210-28, https://doi.org/10.31696/2072-5795-2026-2-5-11.
60. Tepkeev, Vladimir Toltaevich. "Kalmytsko-Dzhungarskie Otnosheniya i Kochevya Kalmytskogo Khanstva v Nizhnem Povolzhe v Kontse XVII Veka [Kalmyk-Dzungar Relations and the Nomadic Pastures of the Kalmyk Khanate on the Lower Volga in the Late 17th Century]." *Oriental Studies*, vol. 18, no. 1, 2025, pp. 52-66, https://doi.org/10.54770/20729286-2025-2-257.
61. Prudnikov, Yaroslav Vladimirovich. "Donskoy i Donetskiy Kazachiy Gorodok XVI-XVII Vv. [Don and Donets Cossack Fortified Settlements in the 16th-17th Centuries]." *Izvestiya Vuzov: Severo-Kavkazskiy Region*, no. 2, 2024, pp. 38-49, https://doi.org/10.15688/jvolsu4.2024.2.4.
62. Albogachiev, Magomed Mikhaylovich. "Akkinskie Migratsii i Rasselenie Vakhnakhov v XV-XVII Vv. [Vainakh and Akkin Migrations and Piedmont Resettlement in the 15th-17th Centuries]." *Vestnik Instituta Istorii, Arkheologii i Etnografii DNTs RAN*, vol. 21, no. 1, 2025, pp. 74-89, https://doi.org/10.7256/2454-0609.2025.2.70633.

63. Dadaev, Yusup Usmanovich. “Sistema Organov Vlasti i Upravleniya Feodalnykh Vladeniy i Soyuzov Selskikh Obshchestv Dagestana v XV-XVIII Vv. [System of Governance in Feudal Polities and Unions of Rural Jamaats of Dagestan in the 15th-18th Centuries].” *Vestnik Dagestanskogo Nauchnogo Tsentra*, no. 53, 2014, pp. 82-91, https://cyberleninka.ru/article/n/sistema-organov-vlasti-i-upravleniya-feodalnyh-vladeniy-dagestana.
64. Mustafina, Dina Abdulbarovna. “Materialy Opisaniy 1685-1687 Gg. Kak Istochnik Po Istorii Zemlevladeniya v Kazanskom Uezde [Materials of the 1685-1687 Surveys as a Source on Landholding in Kazan Uezd].” *Uchenye Zapiski Kazanskogo Universiteta*, vol. 150, no. 1, 2008, pp. 88-99, https://cyberleninka.ru/article/n/materialy-opisaniy-1685-1687-gg-kak-istochnik-po-istorii-zemlevladeniya-v-kazanskom-uezde.
65. Belyakov, Andrey Vasilevich. “Nevidimki Russkoy Armii XVI Veka [Service Tatars of Kasimov, Kadom and Temnikov in the 16th-Century Russian Army].” *Zolotoordynskoe Obozrenie*, no. 2, 2013, pp. 145-62, https://cyberleninka.ru/article/n/nevidimki-russkoy-armii-xvi-veka.
66. Belyakov, Andrey Vasilevich. “Temnikovskoe Knyazhestvo Po Istochnikam XVI-XVII Vv. [The Temnikov Principality According to 16th-17th Century Sources].” *Srednevekovye Tyurko-Tatarskie Gosudarstva*, no. 6, 2014, pp. 31-44, https://cyberleninka.ru/article/n/temnikovskoe-knyazhestvo-po-istochnikam-xvi-xvii-vv.
67. Rezun, Dmitriy Yakovlevich, and Ruslan Sergeevich Vasilievsky. *Letopis Sibirskikh Gorodov [Chronicle of Siberian Towns]*. Novosibirskoe Knizhnoe Izdatelstvo, 1989, https://search.worldcat.org/search?q=Chronicle+Siberian+Towns+Rezun+1989.
68. Matveev, Andrey Vasilevich. “Puti Soobshcheniya Sibirskikh Khanstv [Overland and Riverine Communication Routes of the Siberian Khanates in the 15th-16th Centuries].” *Vestnik Omskogo Universiteta*, no. 3, 2011, pp. 181-90, https://cyberleninka.ru/article/n/puti-soobscheniya-sibirskih-hanstv.
69. Vizgalov, Georgiy Petrovich, and Oleg Viktorovich Kardash. “Ostyatskaya Usadba v Posade Goroda Berezova XVII-XVIII Vv. [Indigenous and Russian Urban Households in the Posad of Berezov in the 17th-18th Centuries].” *Arkheologiya, Etnografiya i Antropologiya Evrazii*, no. 2, 2011, pp. 94-105, https://cyberleninka.ru/article/n/ostyatskaya-usadba-v-posade-goroda-berezova-xviii-v.
70. Solodkin, Yankel Gutmanovich. “Kodskoe Knyazhestvo i Russkie Vlasti Sibiri v Kontse XVI - Pervoy Treti XVII v. [The Koda Khanty Principality and Russian Authorities in Siberia in the Late 16th and Early 17th Centuries].” *Vestnik Ugrovedeniya*, no. 3, 2015, pp. 112-21, https://cyberleninka.ru/article/n/kodskoe-knyazhestvo-i-russkie-vlasti-sibiri-v-kontse-xvi-nachale-xvii-vv.
71. Solodkin, Yankel Gutmanovich. “Pervye Novokreshcheny Severo-Zapadnoy Sibiri (Konets XVI - Pervaya Tret XVII v.) [the First Baptised Indigenous Inhabitants and Garrison Settlements of North-Western Siberia].” *Vestnik Tomskogo Gosudarstvennogo Universiteta: Istoriya*, no. 58, 2019, pp. 25-31, https://cyberleninka.ru/article/n/pervye-novokrescheny-severo-zapadnoy-sibiri-konets-xvi-pervaya-tret-xvii-v.
72. Tychinskikh, Zaytuna Aptrashitovna. “Istoriko-Demograficheskaya Kharakteristika Tatarskogo Naseleniya Tobolskogo i Tyumenskogo Uezdov v XVII-XVIII Vv. [Historical-Demographic Profile of the Tatar Population of Tobolsk and Tyumen Uezds in the 17th-18th Centuries].” *Zolotoordynskoe Obozrenie*, vol. 7, no. 2, 2019, pp. 342-61, https://doi.org/10.17223/15617793/447/22.
73. Kolesnikov, Aleksandr Dmitrievich. *Russkoe Naselenie Zapadnoy Sibiri v XVIII - Nachale XIX Vv. [The Russian Population of Western Siberia in the 18th and Early 19th Centuries]*. Zapadno-Sibirskoe Knizhnoe Izdatelstvo, 1973, https://search.worldcat.org/search?q=Russian+Population+Western+Siberia+18th+Early+19th+Centuries+Kolesnikov+1973.
74. Kabuzan, Vladimir Maksimovich. *Narody Rossii v XVIII Veke: Chislennost i Etnicheskiy Sostav [Peoples of Russia in the 18th Century: Population Size and Ethnic Composition]*. Nauka, 1990, https://search.worldca

t.org/search?q=Peoples+Russia+18th+Century+Population+Size+Ethnic+Composition+Kabuzan+1990.

75. Turaev, Vadim Anatolevich. “Rossiysko-Kitayskoe Pogranichye v XVII Veke [the Russian-Chinese Borderland in the Amur Basin in the 17th Century: Albazin, Nerchinsk and Daur-Ducher Demography].” *Rossiya i ATR*, no. 2, 2020, pp. 31-48, https://doi.org/10.24411/2658-5960-2020-10022.
76. Turaev, Vadim Anatolevich. “Khozhdenie Vstrech Solntsu v Kontekste Problem Prisoedineniya Dalnego Vostoka k Rossiyskomu Gosudarstvu (XVII-XVIII Vv.) [Moving Towards the Sun: Russian Annexation and Indigenous Demography of the Far East in the 17th-18th Centuries].” *Rossiya i ATR*, no. 1, 2013, pp. 5-21, https://cyberleninka.ru/article/n/hozhdenie-vstrech-solntsu-v-kontekste-problem-prisoedineniya-dalnego-vostoka-k-rossiyskomu-gosudarstvu-hvii-hviii-vv.
77. Ushnitskiy, Vasiliy Vasilevich. “Yakutskie Volosti v 30-40-Kh Godakh XVII Veka [Yakut Volosts and Yasak Assessments in the 1630s-1640s].” *Severo-Vostochnyy Gumanitarnyy Vestnik*, no. 3, 2017, pp. 26-34, https://cyberleninka.ru/article/n/yakutskie-volosti-v-30-40-h-godah-xvii-veka.
78. Tolkacheva, Natalya Vladimirovna. “Russkie Lyudi i Korennye Narody Kamchatki: Pervye Kontakty (Konets XVII - Nachalo XVIII v.) [Russian Servicemen and Indigenous Itelmen and Koryak Peoples of Kamchatka at the Turn of the 17th-18th Centuries].” *Vestnik Kemerovskogo Gosudarstvennogo Universiteta*, no. 4, 2015, pp. 85-92, https://doi.org/10.17217/2079-0333-2015-32-100-109.